

# KODE Printer System Architecture and Engineering Reference

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KODE SPORTS CLUB

T E C H N O L O G Y D E P A R T M E N T

KODE Printer  
System Architecture and  
Engineering Reference

|                |                                                           |
|----------------|-----------------------------------------------------------|
| DOCUMENT REF   | KODE-TECH-0005                                            |
| VERSION        | v2.0 · Phase 1 Architecture Baseline                      |
| DATE           | 14 August 2026                                            |
| SUPERSEDES     | KODE-TECH-0004 (v2.0 As-Built Review)                     |
| PREPARED FOR   | Technology / IT Leadership and the implementing engineers |
| CLASSIFICATION | Internal, Confidential                                    |

Document Control

This document is the single source of truth for the KODE Printer system. It describes the system as it exists today and the target architecture it is being built toward, in one place, with every capability explicitly tagged so the difference between the two is never ambiguous.

It supersedes KODE-TECH-0004 (v2.0, As-Built Review), which is retained as a frozen point-in-time record. Where KODE-TECH-0004 documented only what had been built, this document adds the target design, the normative specification an engineer or an AI coding agent needs to implement it, and the decision record behind every significant choice.

| Field             | Value                                                                                      |
|-------------------|--------------------------------------------------------------------------------------------|
| Document Ref      | KODE-TECH-0005                                                                             |
| Version / Status  | v1.0 · Living Document                                                                     |
| Date              | 11 August 2026                                                                             |
| Prepared For      | Technology / IT Leadership and the implementing engineers                                  |
| Department        | Technology                                                                                 |
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| Approver          | To be assigned                                                                             |
| Next Review       | 11 November 2026                                                                           |
| Supersedes        | KODE-TECH-0004 (v2.0 As-Built Review, 11 Aug 2026)                                         |
| Related Documents | KODE-TECH-0002 (PRD) · KODE-TECH-0003 (v1.0 Architecture) · KODE-TECH-0004 (v2.0 As-Built) |
| Reviewed Artifact | backend-ready-fixed (Node.js / Express + PostgreSQL)                                       |
| Canonical Format  | Markdown. This PDF is generated from it and is never edited directly.                      |
| Classification    | Internal, Confidential                                                                     |

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0. How to Read This Document

PART A: NARRATIVE

A1. Executive Summary

KODE Printer provides centralized control and an auditable record of application printing and printer/device activity, without installing software or drivers on user devices. Users upload a file in a browser, choose a permitted printer, configure normal print options, and submit the job. Every application job is logged. Activity started directly at a printer's own panel is detected where the device exposes usable counters.

The Phase 1 target combines the strongest technical decisions from the previous architecture with a deliberately smaller deployment model and a first-class Client Print Experience.

1. IPP is the primary print transport, with RAW/9100 retained as fallback.
2. A durable PostgreSQL queue makes jobs survive restarts and supports controlled transient retries.
3. Printer location is a simple zone attribute; serial number is the physical identity anchor. No site/floor/room hierarchy is required.
4. Notifications and administrative changes are persisted and auditable. Walk-up attribution uses an outstanding-impression ledger, and reports use "Device activity" when a device cannot distinguish printing from copying/faxing.
5. The user flow is explicit: upload -> preview -> choose authorized printer -> choose options -> print -> clear progress/result. Queue, transport, SNMP and audit mechanics remain invisible to ordinary users.

The system is a modular monolith with PostgreSQL. Redis, a site collector, scan management, quotas and enterprise identity features are outside Phase 1.

A2. Scope and Non-Goals

A2.1 In scope

- Browser-based upload and print to authorized network printers, with no client-side software, driver or desktop agent
- Normal print workflow: upload, preview, printer selection, copies, colour/grayscale, duplex, page selection, confirmation, and clear progress/failure feedback
- Per-user, per-printer access control
- Durable print-job tracking and an exportable administrative record
- Printer administration: add, edit, disable, probe, monitor and assign access

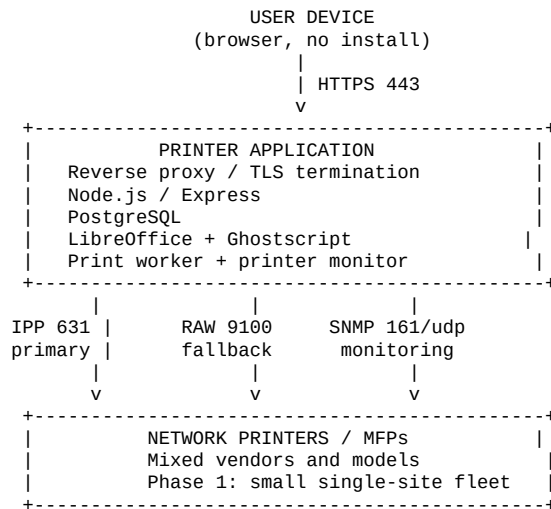
- Simple printer zones for discovery, filtering and reporting
- Mixed printer vendors and models using IPP where supported and RAW/9100 as fallback
- Walk-up/device activity monitoring through printer counters where usable telemetry exists
- Administrative reporting, notifications and append-only audit history

#### A2.2 Explicitly out of scope for Phase 1

- Site collector / distributed collector deployment
- Building/floor/room facility hierarchy
- Print release / follow-me printing
- Mobile-native applications
- Desktop print agents or OS print-driver installation
- Scan management and scan-to-folder ingestion
- Quotas, chargeback, billing and cost enforcement
- MFA and enterprise identity federation
- Redis, RabbitMQ, Kafka or another external queue service
- Public internet exposure and multi-tenancy

These are deliberate Phase 1 boundaries. They do not prevent later extensions; they prevent infrastructure and business logic that has no current consumer from becoming part of the core product.

#### A3. System Context

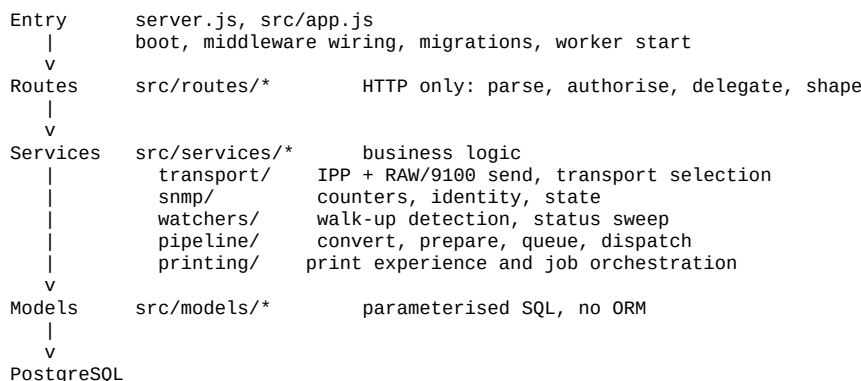


The application server is the only component that communicates with printers. The browser never opens printer connections directly. The Phase 1 deployment profile is approximately 12 network printers distributed across roughly 3-4 logical zones, with no assumption that all devices expose identical capabilities.

The architecture is vendor-neutral. Printer differences are handled behind transport, capability and monitoring interfaces. Untrusted inputs include uploaded documents, SNMP/device responses, and printer-reported capability/state data.

#### A4. Architecture at a Glance

Five logical layers remain, with collector and scan subsystems removed from the Phase 1 target:



Supporting layers: src/config/\* (the only place process.env is read), src/middlewares/\* (auth, validation, upload filtering, rate limiting, security headers and error handling), and src/utilities/\* (stateless helpers).

Background work is split logically into a print worker and a printer-monitor worker, but both remain part of the same deployable application. No external queue service is required.

#### A5. Key Decisions Summary

- A6. Network and Zone Strategy
- A7. PRD Coverage and Traceability
- A8. Risk Register

## A9. Backlog

Every Phase 1 requirement and remaining gap is prioritised here. Items removed from the previous v1.0 target because they were deployment-scale or unrelated to printing are intentionally not carried as Phase 1 blockers.

### A9.1 Blocking go-live

|        |                                               |       |
|--------|-----------------------------------------------|-------|
| GAP-01 | Default admin credentials and forced rotation | B12.4 |
| GAP-02 | TLS termination and secure production binding | B18.3 |
| GAP-03 | Process supervision, restart and log rotation | B18.4 |
| GAP-04 | Default database credentials blocked at boot  | B3.3  |
| GAP-05 | Clean production upload/storage directories   | B19.1 |
| GAP-06 | Converter prerequisites verified at boot      | B10.6 |
| GAP-07 | Tested backup and restore procedure           | B19.4 |

### A9.2 Phase 1 correctness and trust

|        |                                                                             |               |
|--------|-----------------------------------------------------------------------------|---------------|
| GAP-08 | Walk-up detection separates application prints from generic device activity | B8.4          |
| GAP-09 | Outstanding-impression ledger replaces fixed time-window attribution        | B8.3          |
| GAP-10 | Append-only admin audit log                                                 | B4.9          |
| GAP-11 | Persisted notifications with per-admin read state                           | B4.8          |
| GAP-12 | Durable PostgreSQL queue and controlled retries                             | B10.4         |
| GAP-13 | Serial-number printer identity                                              | B7.1          |
| GAP-14 | Per-model capability detection                                              | B7.3          |
| GAP-15 | Printer status beyond reachability where IPP/SNMP supports it               | B7.4          |
| GAP-16 | Maintained SNMP library instead of hand-rolled BER parsing                  | B8.1          |
| GAP-17 | Converter isolation, timeouts and per-job working directories               | B10.3 / B16.2 |
| GAP-18 | Authentication hardening and account protection                             | B12.4         |
| GAP-19 | Bounded job/report queries                                                  | B5.4          |
| GAP-20 | Structured logging, health/readiness and basic alerting                     | B13           |
| GAP-21 | Automated tests for critical print/authorization/measurement paths          | B17           |

### A9.3 Phase 1 user experience

|       |                                                                      |      |
|-------|----------------------------------------------------------------------|------|
| UX-01 | Upload -> preview -> print workflow                                  | B5.5 |
| UX-02 | Printer picker shows only authorized printers and useful status      | B5.5 |
| UX-03 | Copies, colour/grayscale, duplex and page selection remain available | B5.5 |
| UX-04 | Clear progress and final job state without exposing queue internals  | B5.5 |
| UX-05 | User-friendly mapping of technical failures                          | B5.5 |
| UX-06 | Remember last/default printer where appropriate                      | B5.5 |

### A9.4 Deferred / removed from Phase 1

|          |                                                           |      |
|----------|-----------------------------------------------------------|------|
| DEFER-01 | Site collector and segmented-network protocol             | B11  |
| DEFER-02 | sites / floor / room facility hierarchy                   | B4   |
| DEFER-03 | Scan detection and scan-share ingestion                   | B9   |
| DEFER-04 | Quotas, chargeback and cost enforcement                   | B4   |
| DEFER-05 | MFA / enterprise identity federation                      | B12  |
| DEFER-06 | Prometheus/Zabbix-scale monitoring stack                  | B13  |
| DEFER-07 | Advanced keyset pagination as a universal API requirement | B5.4 |

These items may become valid future capabilities. Their absence does not block Phase 1 printing.

## A10. Roadmap

| Release          | Theme                     | Contents                                                                                                      | Exit criterion                                                                     |
|------------------|---------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Phase 1          | Normal printing + control | Print UX, permissions, IPP/RAW, queue, monitoring, device activity, audit, notifications, reporting, security | Users can print without losing normal print controls; admins can trust the record. |
| Phase 1.x gains. | Reliability refinement    | Additional vendor capability probes, monitoring coverage and operational hardening                            | No change to the core UX; measurable reliability                                   |
| Future it.       | Distributed deployment    | Optional collector for genuinely segmented networks and multi-site requirements                               | Introduced only when a real network boundary requires                              |
| Future exists.   | Governance / enterprise   | Quotas, cost reporting, MFA, federation, advanced observability                                               | Added only when a product or deployment requirement                                |

The Phase 1 architecture is intentionally complete for the stated deployment profile. Future scale is handled by explicit extensions rather than by shipping infrastructure that has no current consumer.

## A11. Decisions Required

| ID     | Decision                                 | Recommendation                                                                         |
|--------|------------------------------------------|----------------------------------------------------------------------------------------|
| DEC-01 | Named document owner and approver        | Assign before implementation sign-off.                                                 |
| DEC-02 | Exact printer inventory and capabilities | Probe every printer before production cutover.                                         |
| DEC-03 | Uploaded file retention                  | Default 30 days, then purge; retain job metadata.                                      |
| DEC-04 | Walk-up terminology                      | Use "Device activity" unless the device exposes a reliable print/copy/fax distinction. |
| DEC-05 | Backup target and retention              | Nightly backup to the existing backup target; test restore.                            |

|        |                         |                                                                                      |
|--------|-------------------------|--------------------------------------------------------------------------------------|
| DEC-06 | Default/last printer UX | Remember the user's last successful printer; do not let this override authorization. |
|--------|-------------------------|--------------------------------------------------------------------------------------|

No decision is required to introduce a collector, site hierarchy, quotas or MFA in Phase 1.

#### A12. Change Log

| Version | Date           | Author                | Change                                                                                                                                                                                                                                                                                                                                  |
|---------|----------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v2.0    | 14 August 2026 | Technology Department | Phase 1 architecture baseline. Retains the v1.0 reliability, security, IPP, queue, printer identity, monitoring and audit decisions; replaces multi-site/site-collector assumptions with a simple zone model; adds the Client Print Experience; removes scan management, quotas and other non-Phase-1 complexity from the target scope. |
| v1.0    | 11 August 2026 | Technology Department | Initial issue of KODE-TECH-0005. Supersedes KODE-TECH-0004. Added target architecture, normative specification, ADRs, multi-site design and corrected walk-up measurement model.                                                                                                                                                        |

#### PART B: NORMATIVE ENGINEERING REFERENCE

- B1. Invariants
- B2. Repository Layout

Target layout for Phase 1. The delivered repository may differ until the migration work is complete.

```
src/
  routes/      auth, printers, jobs, users, stats, notifications, health
  services/    auth, printers, jobs, printing, monitoring, reports
  models/      one module per table, parameterised SQL
  transport/   ipp, raw9100, selection
  snmp/        identity, counters, status
  pipeline/    validation, conversion, preparation, queue, dispatch
  watchers/    pageCounter, status, scheduler
  config/      environment/configuration access
  middlewares/ auth, validation, upload, rate limiting, security
  utilities/   stateless helpers

workers/      print worker and printer monitor worker
```

There is no collector module, scan-folder watcher, sites CRUD module or facility hierarchy in Phase 1.

#### B3. Configuration Reference

- B4. Data Model
- B5. API Contract
- B5.5 Client Print Experience

The client experience is part of the Phase 1 system contract. The goal is that a user can print normally without understanding queues, workers, transports, SNMP or audit records.

Required flow:

```
Upload -> validate -> prepare -> preview -> choose authorized printer -> choose print options -> confirm
-> 202 Accepted -> progress -> completed / failed
```

The UI MUST support copies, colour/grayscale, duplex and page selection. Office files are converted to the final printable representation before page selection is applied so that page numbers correspond to what will actually be printed.

The printer picker MUST show only printers authorized through user\_printers. Zone and recent-printer information may improve ordering, but neither may grant access. The UI SHOULD remember the last successfully used printer.

Printer status is displayed as Available, Warning, Unavailable or Unknown. Unknown means the system cannot confidently determine the state; it MUST NOT be presented as healthy.

After submission, the browser polls GET /jobs/:id at a modest interval. Internal states such as queued, processing and sending are translated into simple user-facing progress. Technical error codes remain in the backend record; users receive actionable messages such as "This printer is unavailable" or "This file type is not supported."

The UI MUST NOT expose queue internals, worker IDs, SNMP details, audit terminology or transport implementation details to ordinary users.

#### B5.5 Client Print Experience

The client experience is part of the Phase 1 system contract. The goal is that a user can print normally without understanding queues, workers, transports, SNMP or audit records.

Required flow:

```
Upload -> validate -> prepare -> preview -> choose authorized printer -> choose print options -> confirm
-> 202 Accepted -> progress -> completed / failed
```

The UI MUST support copies, colour/grayscale, duplex and page selection. Office files are converted to the final printable representation before page selection is applied so that page numbers correspond to what will actually be printed. The printer picker MUST show only printers authorized through user\_printers. Zone and recent-printer information may

improve ordering, but neither may grant access. The UI SHOULD remember the last successfully used printer.

Printer status is displayed as Available, Warning, Unavailable or Unknown. Unknown MUST NOT be presented as healthy. After submission, the browser polls GET /jobs/:id at a modest interval and translates internal states into simple user-facing progress. Technical error codes remain in the backend; users receive actionable messages. Queue internals, worker IDs, SNMP details, audit terminology and transport implementation details are not exposed to ordinary users.

#### B6. Printer Transport

- B7. Printer Identity, Capability and State
- B8. Walk-Up Job Detection
- B9. Scan Detection – OUT OF SCOPE FOR PHASE 1
- B10. Print Pipeline and Job Queue
- B11. Distributed Network Extension – Deferred
- B12. Authentication and Authorisation
- B13. Observability
- B14. Failure Modes
- B15. Non-Functional Requirements
- B16. Threat Model
- B17. Test Strategy
- B18. Deployment Topology
- B19. Operations Runbook
- B20. Architecture Decision Records

#### 0. How to Read This Document

##### PART A: NARRATIVE

- A1. Executive Summary
- A2. Scope and Non-Goals
- A3. System Context
- A4. Architecture at a Glance
- A5. Key Decisions Summary

## A6. Network and Zone Strategy

Phase 1 assumes a single logical deployment in which the application server can reach the configured printers. Physical placement is represented by a simple printer.zone value. Zone is a user-facing grouping and reporting attribute, not a facility-management subsystem.

### A6.1 Zone model

printers.zone -> simple text or small lookup value

Example:

```
Printer 01 -> Zone A
Printer 02 -> Zone A
Printer 03 -> Zone A
Printer 04 -> Zone B
Printer 05 -> Zone C
```

The UI may prefer recent or nearby-zone printers, but authorization remains based only on user\_printers. Zone MUST NOT grant permission.

### A6.2 Network reachability

For each configured printer, the server uses the protocols actually supported by that device: TCP 631 for IPP where available, TCP 9100 for RAW fallback where available, and UDP 161 for SNMP monitoring where enabled.

A failed protocol probe does not automatically create another deployment component. The printer is recorded with the capabilities and monitoring coverage that are actually available.

### A6.3 Future segmented networks

A site collector is deliberately deferred. If a later deployment proves that a printer segment cannot be reached directly and policy forbids the required inbound protocols, a collector can be introduced as a deployment extension. It is not part of the Phase 1 code path, data model, API surface or operating requirements.

## A7. PRD Coverage and Traceability

- A8. Risk Register
- A9. Backlog
- A10. Roadmap
- A11. Decisions Required
- A12. Change Log

## PART B: NORMATIVE ENGINEERING REFERENCE

- B1. Invariants
- B2. Repository Layout
- B3. Configuration Reference
- B4. Data Model
- B5. API Contract
- B6. Printer Transport
- B7. Printer Identity, Capability and State
- B8. Walk-Up Job Detection
- B9. Scan Detection – OUT OF SCOPE FOR PHASE 1

The previous architecture included scan-to-folder ingestion and file-stability handling. Those mechanisms are not part of the Phase 1 printer application because the product requirement is printing, printer monitoring and print activity accounting. Scan management may be added later as an independent capability.

No scan watcher, scan share, scan job type, scan ingestion API or scan-specific retention policy is required for Phase 1.

### B10. Print Pipeline and Job Queue

- B11. Site Collector `PLANNED` (GAP-27)
- B12. Authentication and Authorisation
- B13. Observability `PLANNED` (GAP-21)
- B14. Failure Modes
- B15. Non-Functional Requirements
- B16. Threat Model
- B17. Test Strategy `PLANNED` (GAP-22)
- B18. Deployment Topology
- B19. Operations Runbook
- B20. Architecture Decision Records

0. How to Read This Document

0.1 Two parts, two audiences

Part A (Sections A1 to A12) is narrative. It is written for Technology and IT leadership. It explains what the system does, what it costs, what the risks are, and what decisions are outstanding. Read it end to end.

Part B (Sections B1 to B20) is normative. It is written for the implementing engineer and for any AI coding agent given this document as reference. It contains exact schemas, contracts, algorithms, and rules. Read it as specification, not as description.

Where Part A and Part B disagree, Part B is authoritative. Part A summarises. Part B binds.

0.2 Status tags

Every capability in this document carries exactly one tag:

| Tag     | Meaning                                                                               |
|---------|---------------------------------------------------------------------------------------|
| BUILT   | Implemented and present in the reviewed source. Read the code before changing it.     |
| PARTIAL | Implemented, with a known edge case, manual step, or scope limit stated alongside it. |
| GAP     | Not implemented and not yet designed. Called out so it is planned, not discovered.    |
| PLANNED | Designed in this document, not yet written. This is code to be created.               |

An engineer or agent implementing from this document should treat PLANNED as the work queue and BUILT as the constraint set.

0.3 Requirement language

MUST and MUST NOT are absolute. Violating one is a defect regardless of whether a test catches it. SHOULD and SHOULD NOT indicate a strong default. Deviating requires a stated reason recorded in the change log. MAY indicates a genuine option with no preferred answer.

0.4 Glossary

| Term                | Meaning                                                                                                                     |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| IPP attributes and  | Internet Printing Protocol. HTTP-based standard printing protocol, TCP 631. Carries job returns real printer and job state. |
| IPP Everywhere      | Certification requiring driverless IPP support and application/pdf acceptance. Underpins AirPrint and Mopria.               |
| RAW / 9100 feedback | JetDirect raw printing. A plain TCP socket on port 9100 into which file bytes are written. No channel.                      |

| Term               | Meaning                                                                                                                                                            |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PJL<br>colour mode | Printer Job Language. HP-originated text directives prefixed to a job to set copies, duplex, and paper size without a driver. Widely but not universally honoured. |
| PCL                | Printer Command Language. HP page description language family.                                                                                                     |
| SNMP               | Simple Network Management Protocol, UDP 161. Used here to read page counters and device identity.                                                                  |
| MIB / OID          | Management Information Base and Object Identifier. The named tree and the numeric address of a value readable over SNMP.                                           |
| BER / ASN.1        | The binary encoding rules SNMP messages use on the wire.                                                                                                           |
| Printer-MIB        | RFC 3805 standard MIB implemented by most network printers. Source of the page counter.                                                                            |
| MFP                | Multi-Function Peripheral. A device that prints, scans, copies and often faxes.                                                                                    |
| Walk-up job        | A job started at the device's own control panel, outside this system.                                                                                              |
| Impression<br>the  | One side of one sheet marked by the print engine. A duplex page is two impressions. This is what page counter counts.                                              |
| Collector          | An optional instance of this system running in reduced mode inside a network segment the central server cannot reach directly.                                     |

## PART A: NARRATIVE

## A1. Executive Summary

KODE Printer gives the club central control and a complete audit trail over printing and scanning across the facility, without installing software or drivers on any staff device. Users upload a file in a browser, choose a printer they are permitted to use, and the server sends it. Every job is logged. Jobs started directly at a printer's own panel are detected and logged too, so the record is complete rather than only covering what went through the app.

The delivered build works. It prints, converts Office documents, enforces per-printer permissions, tracks walk-up jobs, and exports reports. It is not yet production ready: default credentials ship enabled, TLS termination is not configured, and there is no process supervision. Those are the items blocking go-live and they are small.

The target architecture in this document makes four substantive changes to the delivered design:

1. Adopt IPP as the primary print transport, with RAW/9100 retained as fallback. This replaces guesswork with a standards-based feedback channel and closes two known limitations outright: printer state (toner, paper, jams) and reliable per-model support for copies, duplex and colour.
2. Introduce a durable job queue. Not for throughput, which is low, but so that a job survives a server restart, retries automatically when a printer is mid-reboot, and does not die silently behind a browser timeout during a long conversion.
3. Model sites, buildings and printer identity properly. The club occupies multiple buildings. The current schema has no location field and identifies printers solely by IP address, which breaks the moment a DHCP lease changes.
4. Make the record trustworthy. Persist notifications, add an admin audit log, and correct a measurement flaw in walk-up detection (see A7) that currently counts photocopies as prints.

None of these require abandoning the existing codebase. All are additive.

## A2. Scope and Non-Goals

## A2.1 In scope

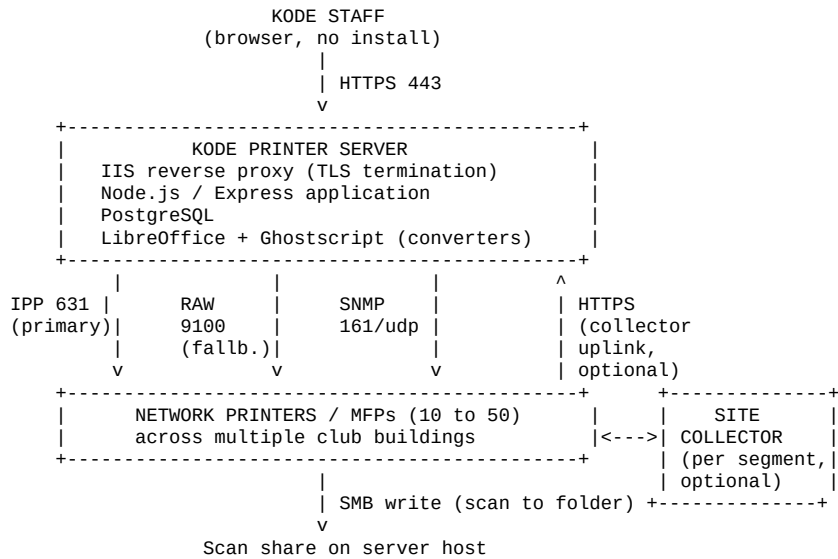
- Browser-based upload and print to any networked printer in the facility, with no client-side software or driver installation
- Per-user, per-printer access control
- A complete, exportable audit trail of print and scan activity, including activity started at the device
- Printer fleet administration: add, edit, disable, monitor
- Operation across multiple buildings on the club's network

A2.2 Explicitly out of scope

Recording these prevents scope drift and sets expectations with stakeholders:

- Print release / follow-me printing (badge in at any device to release a held job). Valuable, materially larger, deferred beyond v2.0.
- Mobile-native applications. The web interface is responsive; there is no iOS or Android app.
- Consumables procurement. The system reports toner state (v1.1); it does not order supplies.
- Chargeback and invoicing. Page quotas and cost reporting are planned for v1.1 as reporting only, not as a billing integration.
- Public internet exposure. This is an internal system on the club network. See A6 and B18.
- Multi-tenancy. One club, one deployment.

A3. System Context

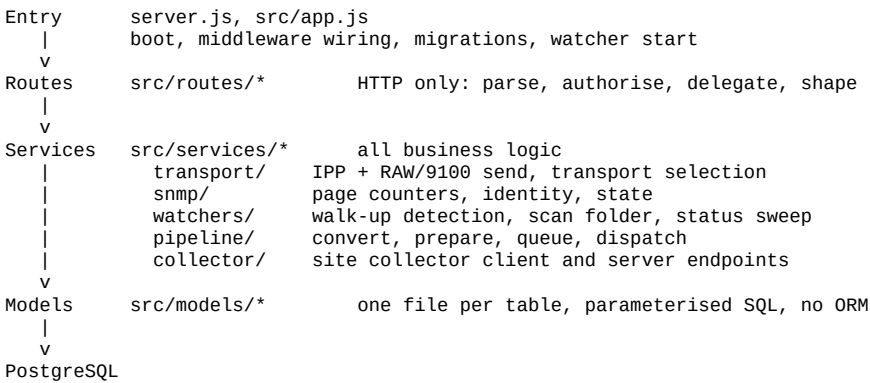


Three trust boundaries matter and each is treated as hostile input in the threat model (B16):

1. Files uploaded by staff, which are parsed by LibreOffice and Ghostscript
2. SNMP replies arriving over unauthenticated UDP
3. Files written into the scan share by printer firmware

A4. Architecture at a Glance

Five layers in the backend. Nothing reaches across a layer boundary: a route never opens a socket, a service never reads process.env .



Supporting layers: src/config/\* (the only place process.env is read), src/middlewares/\* (auth, validation, upload filtering, rate limiting, security headers, error handling), src/utilities/\* (stateless helpers).

Each protocol's byte-level code stays isolated in its own subfolder behind a single stable export. Swapping the SNMP implementation or adding a transport touches one folder, not every caller.

A5. Key Decisions Summary

Full context, alternatives and consequences for each are in Part B, Section B20.

| #                     | Decision                                                                                      | Status                                    |
|-----------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------|
| ADR-001               | IPP is the primary print transport; RAW/9100 is the fallback                                  | Accepted, PLANNED                         |
| ADR-002               | Reject the Windows print spooler and Windows Print Server                                     | Accepted                                  |
| ADR-003<br>department | Keep Node.js, Express and pg ; adopt TypeScript incrementally;<br>no NestJS or Prisma rewrite | Accepted, deviation from<br>default stack |
| ADR-004               | Durable print queue in PostgreSQL using FOR UPDATE SKIP<br>LOCKED , not Redis                 | Accepted, PLANNED                         |
| ADR-005               | Replace the hand-rolled SNMP client with a maintained library;<br>prefer SNMPv2c, support v3  | Accepted, PLANNED                         |
| ADR-006               | Printer identity anchors on serial number; IP address is mutable<br>metadata                  | Accepted, PLANNED                         |
| ADR-007               | Optional site collector for network segments the server cannot<br>reach                       | Accepted, PLANNED                         |
| ADR-008               | Walk-up detection uses an outstanding-impression ledger, not a<br>fixed time window           | Accepted, PLANNED                         |
| ADR-009               | Scan-to-folder retained, with a file-stability gate before ingestion                          | Accepted, PLANNED                         |

| #       | Decision                                                            | Status            |
|---------|---------------------------------------------------------------------|-------------------|
| ADR-010 | Document converters run sandboxed, unprivileged, with hard timeouts | Accepted, PLANNED |
| ADR-011 | Notifications persist to PostgreSQL with per-admin read state       | Accepted, PLANNED |
| ADR-012 | Every administrative mutation writes to an append-only audit log    | Accepted, PLANNED |

#### A6. Multi-Site Strategy

The club occupies several buildings. The network layout between them is not yet confirmed, and the architecture is deliberately designed so that the answer does not change the codebase.

##### A6.1 The two possibilities

Flat network. Every printer is routable from the server on TCP 631, TCP 9100 and UDP 161, and every printer can write to the scan share. The server talks to every device directly and no collector is deployed. This is the simple case.

Segmented network. Buildings sit on separate VLANs with filtering between them. Opening three inbound protocols per building, including unauthenticated UDP and SMB, is a request most security policies will refuse, and rightly so.

##### A6.2 The design that covers both

A site collector is the same application running in reduced mode: it polls SNMP, watches the local scan share, and sends print jobs for printers in its own segment only. It reports to the central server over a single outbound HTTPS connection. The central server holds all state, all users and all reporting.

On a flat network, zero collectors are deployed and nothing about the system changes. On a segmented network, one collector per building replaces three inbound firewall exceptions per building with one outbound HTTPS connection.

The cost of designing this boundary now is a handful of interfaces. The cost of retrofitting it later is a rewrite of every service that touches a printer.

##### A6.3 Resolving the unknown, this week

Do not wait for a VLAN map. From the application server, against one printer in each building, attempt in order:

1. Test-NetConnection <printer-ip> -Port 631
2. Test-NetConnection <printer-ip> -Port 9100
3. An SNMP get of 1.3.6.1.2.1.1.5.0 ( sysName )
4. From the printer's own web interface, a test write to the scan share

Any failure identifies a real boundary. Record the results in the per-printer inventory (B19.2). Request the VLAN map in parallel, but let the test drive the decision.

## A6.4 What multiple buildings changes in the data model

- Printers gain a site. sites becomes a first-class table; every printer belongs to one and carries floor and room. Without this, an admin cannot answer "which printer is near me" and no report can be broken out by building. GAP in the delivered schema.
- IP address stops being identity. The delivered schema makes ip\_address unique and treats it as the printer's identity. A changed DHCP lease silently turns a working printer into an unreachable one, and re-adding it splits the audit trail across two records. Identity moves to the SNMP serial number. GAP in the delivered schema.
- Polling is staggered and adaptive. A four second sweep across inter-building links needs jitter and exponential backoff for devices that consistently time out, so one unreachable building does not consume the poll budget.
- Discovery does not cross subnets. mDNS and broadcast discovery stop at the VLAN boundary. Onboarding remains a per-device, per-site action.

## A7. PRD Coverage and Traceability

Cross-checked against KODE-TECH-0002 Section 6. The full requirement to code to test matrix is in B17.3.

| PRD requirement                                   | Status             | Reference                                                                   |
|---------------------------------------------------|--------------------|-----------------------------------------------------------------------------|
| Accounts and permissions, two roles               | BUILT              | B12                                                                         |
| Add, edit, remove printer by IP only              | BUILT              | B4, B5                                                                      |
| Printer online / offline status                   | PARTIAL to PLANNED | Ping-only today. IPP printer-state-reasons in v1.1 gives real state. B6, B7 |
| Upload and print (PDF, Office, image), no install | BUILT              | B10                                                                         |
| v1.1. Copies, colour, duplex, page selection      | PARTIAL to         | PJL today, unverified per model. IPP job attributes in                      |
|                                                   | PLANNED            | B6                                                                          |
| Log every job sent via the app                    | BUILT              | B4                                                                          |
| Detect walk-up jobs                               | PARTIAL            | B8. See the measurement flaw below.                                         |
| Admin dashboard and exportable report             | BUILT              | B5                                                                          |
| Scan tracking                                     | PARTIAL            | B9. Requires per-printer setup on the device.                               |
| Works across printer brands                       | BUILT              | B6                                                                          |

## A7.1 A measurement flaw leadership should be aware of

Walk-up detection reads prtMarkerLifeCount , the printer's lifetime impression counter. That counter increments for every impression the print engine makes: prints, photocopies, received faxes, and internally generated report pages.

The system therefore currently logs photocopies as walk-up print jobs. On an MFP fleet in a sports club, where reception and administration copy membership forms and ID documents routinely, this is not a rounding error. Any report drawn from walk-up data today overstates printing.

Separating the categories requires vendor-specific private OIDs, which differ by manufacturer (HP, Xerox, Canon and Konica Minolta each expose their own). The per-printer inventory pass (B19.2) captures these where available, and B8.4 specifies the behaviour where they are not. Until then, walk-up totals MUST be labelled "device activity" rather than "prints" in any report shown to stakeholders.

#### A8. Risk Register

| ID                   | Risk                                                             | Likelihood | Impact   | Mitigation                                                                        |
|----------------------|------------------------------------------------------------------|------------|----------|-----------------------------------------------------------------------------------|
| R1<br>boot           | Default credentials reach production                             | Medium     | Critical | Forced password change on first login; refuses default secrets. B12.4, GAP-01     |
| R2<br>non-<br>B18    | Traffic unencrypted on the internal network                      | Medium     | High     | IIS terminates TLS; application refuses local plaintext binding in production.    |
| R3                   | Malicious document exploits LibreOffice or Ghostscript           | Low        | High     | Sandboxed converters, unprivileged account, hard timeouts, resource limits. B16.2 |
| R4<br>B8.4           | Audit trail incomplete or misleading                             | Medium     | High     | Photocopy conflation (A7.1), plus SNMP-disabled devices producing silent gaps.    |
| R5                   | Printer IP changes and the device silently drops out             | High       | Medium   | Serial-anchored identity plus DHCP reservations. ADR-006                          |
| R6                   | Network segmentation blocks SNMP or scan share between buildings | Unknown    | Medium   | Site collector design. A6.2                                                       |
| R7<br>notifications. | Server failure loses in-flight jobs and notifications            | Medium     | Medium   | Durable queue and persisted ADR-004, ADR-011                                      |
| R8<br>not            | Converter dependencies missing on a rebuilt host                 | Medium     | Medium   | Startup health check that fails loudly, silently. B10.6                           |
| R9<br>DEC-03,        | Uploads directory grows without bound                            | High       | Low      | Retention policy plus disk alerting. B13.4                                        |
| R10                  | Single server is a single point of failure                       | Low        | High     | Documented restore procedure and RTO. Accepted for v1.0. B19.4                    |

#### A9. Backlog

Every gap and planned item, prioritised, with a stable ID. This table is the ticket backlog. Acceptance criteria for each are in the section referenced.

## A9.1 Blocking go-live

| ID     | Item                                                                 | Section |
|--------|----------------------------------------------------------------------|---------|
| GAP-01 | Default admin credentials auto-seeded with no forced rotation        | B12.4   |
| GAP-02 | No TLS termination configured in the delivered artifact              | B18.3   |
| GAP-03 | No process manager, auto-restart or log rotation                     | B18.4   |
| GAP-04 | Default database credentials not blocked at boot, unlike JWT secrets | B3.3    |
| GAP-05 | Leftover test uploads present in the delivered archive               | B19.1   |
| GAP-06 | Converter prerequisites unverified on the target host                | B10.6   |
| GAP-07 | No backup or restore procedure                                       | B19.4   |

## A9.2 v1.1, correctness and trust

| ID     | Item                                                                   | Section |
|--------|------------------------------------------------------------------------|---------|
| GAP-08 | Walk-up detection conflates copies and faxes with prints               | B8.4    |
| GAP-09 | Fixed 45 second grace window can misattribute large jobs               | B8.3    |
| GAP-10 | No admin audit log for permission and configuration changes            | B4.9    |
| GAP-11 | Notifications held in memory, lost on restart, no per-admin read state | B4.8    |
| GAP-12 | No durable queue or retry on failed sends                              | B10.4   |
| GAP-13 | Printer identity is IP-based and breaks on lease change                | B7.1    |
| GAP-14 | No site, building, floor or room model                                 | B4.2    |
| GAP-15 | Scan watcher can ingest partially written files                        | B9.2    |
| GAP-16 | Concurrent LibreOffice invocations collide on a shared profile         | B10.3   |
| GAP-17 | No conversion timeouts; a malformed file can hang the converter        | B10.3   |
| GAP-18 | Password reset, account lockout and complexity policy absent           | B12.4   |
| GAP-19 | Job list and CSV export unbounded, no pagination                       | B5.4    |
| GAP-20 | Converters run unsandboxed against untrusted input                     | B16.2   |
| GAP-21 | No structured logging, metrics or alerting                             | B13     |
| GAP-22 | No automated tests of any kind                                         | B17     |

## A9.3 v2.0, capability

| ID<br>Section | Item   |                                                                       |
|---------------|--------|-----------------------------------------------------------------------|
| 2             | GAP-23 | IPP transport not implemented B6.                                     |
| 3             | GAP-24 | No per-model capability detection B7.                                 |
| 4             | GAP-25 | Printer status is reachability only, not consumables or jam state B7. |
| 1             | GAP-26 | Hand-rolled SNMP client parses untrusted UDP B8.                      |
|               | GAP-27 | Site collector not implemented B11                                    |
| 10            | GAP-28 | No page quotas or cost reporting B4.                                  |
| B12.4         | GAP-29 | No MFA                                                                |

## A10. Roadmap

| Release           | Theme             | Contents                                  | Exit criterion                                                                  |
|-------------------|-------------------|-------------------------------------------|---------------------------------------------------------------------------------|
| v1.0              | Go-live           | GAP-01 to GAP-07                          | The system runs supervised, over TLS, with rotated secrets and a tested restore |
| v1.1<br>lost to a | Trust the<br>data | GAP-08 to GAP-22                          | Every number in a report is defensible; no job is restart                       |
| v2.0              | Capability        | GAP-23 to GAP-29                          | Real printer state, verified per-model options, segmented-network support       |
| v2.1+             | Deferred          | Print release, quotas enforcement, mobile | Not scheduled                                                                   |

v1.0 is days of work, not weeks. It is configuration and hardening, not feature development. Nothing in v1.1 should start before v1.0 is closed.

## A11. Decisions Required

These are open and need an owner's answer. Each blocks the section named.

| ID     | Decision                           | Options                        | Blocks       | Recommendation              |
|--------|------------------------------------|--------------------------------|--------------|-----------------------------|
| DEC-01 | Named document owner and approver  | A person, not "the department" | Front matter | Assign before circulation   |
| DEC-02 | Network topology between buildings | Flat / segmented / mixed       | B11, B18.2   | Run the A6.3 test this week |

| ID                    | Decision                                                     | Options                                                    | Blocks       | Recommendation                                                                   |
|-----------------------|--------------------------------------------------------------|------------------------------------------------------------|--------------|----------------------------------------------------------------------------------|
| DEC-03<br>Balances    | Uploaded file retention                                      | Delete on success /<br>retain N days / retain indefinitely | B10.7,<br>R9 | Retain 30 days, then purge.<br>audit against disk.                               |
| DEC-04<br>no<br>not   | Scope of the audit trail                                     | Jobs only / jobs plus<br>admin actions                     | B4.9         | Both. A permissions system with<br>record of permission changes is<br>auditable. |
| DEC-05<br>enforcement | Quota and cost reporting                                     | In v1.1 / deferred / out of<br>scope                       | B4.10        | Reporting only in v1.1, no                                                       |
| DEC-06<br>figure is   | Walk-up label in reports<br>until GAP-08 lands               | "Prints" / "Device<br>activity"                            | A7.1         | "Device activity". The current<br>not prints.                                    |
| DEC-07<br>existing    | Backup target and<br>retention                               | Where and for how long                                     | B19.4        | Nightly dump to the club's<br>backup target, 30 day retention                    |
| DEC-08<br>here.       | Whether frontend<br>architecture belongs in<br>this document | Yes / separate document                                    | Scope        | Separate document, referenced<br>Keeps this one focused.                         |

A12. Change Log

| Version         | Date              | Author                   | Change                                                                                                                                                                                    |
|-----------------|-------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1.0<br>design, | 11 August<br>2026 | Technology<br>Department | Initial issue of KODE-TECH-0005. Supersedes KODE-TECH-0004. Adds<br>target architecture, Part B normative specification, ADRs, multi-site<br>and the corrected walk-up measurement model. |

Prior history sits with the documents themselves: KODE-TECH-0004 v2.0 (11 Aug 2026, as-built review), KODE-TECH-0003 v1.0 (09 Aug 2026, pre-implementation architecture), KODE-TECH-0002 (PRD).

## PART B: NORMATIVE ENGINEERING REFERENCE

## B1. Invariants

These rules are absolute. They exist because each one has already caused, or would cause, a class of defect that is expensive to find. Any change that violates one is a defect regardless of test results.

INV-01. Printer permission is derived from the `user_printers` table and nothing else. No route, service, model or report MUST re-derive access from department, site, or any other field. A previous department-string-matching approach caused two systems to disagree about access; that bug is fixed and MUST NOT be reintroduced.

INV-02. Only the `src/config/*` layer reads `process.env`. Every other file reads the merged Config object.

INV-03. Only `src/services/transport/*` opens a socket to a printer. Routes and models MUST NOT.

INV-04. Every SQL statement is parameterised. String interpolation into SQL is forbidden without exception, including in migrations and CSV export.

INV-05. A job row is never hard-deleted and its `printer_id` is never cascaded away. Printers with job history are soft-disabled, and the database enforces this with `ON DELETE RESTRICT`, not application code alone.

INV-06. Every job row carries `username_snapshot` and `printer_name_snapshot` written at insert time, so the record stays readable after the user or printer record changes or is removed.

INV-07. Every administrative mutation writes an `audit_log` row in the same transaction as the change. If the audit write fails, the change fails.

INV-08. No secret is ever logged, returned in an API response, or committed. This includes SNMP community strings, refresh tokens, and collector API keys.

INV-09. Uploaded file bytes are never passed to a converter or a printer before the magic-byte check in B10.2 passes.

INV-10. All timestamps are stored as `TIMESTAMP TZ` in UTC. Local time exists only in the presentation layer.

INV-11. Every asynchronous route handler is wrapped so a rejected promise returns a structured error rather than terminating the process.

INV-12. Stack traces and driver-level error text never appear in an API response body.

B2. Repository Layout

|                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |      |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| backend/<br>server.js<br>src/<br>app.js<br>config/<br><br>middlewares/<br><br>routes/<br>services/<br>transport/<br>snmp/<br>watchers/<br>pipeline/<br><br>printerAccess.js<br>audit.js<br>models/<br>utilities/<br>migrations/<br>test/<br>unit/ integration/ | process entry, signal handling, graceful shutdown<br><br>express assembly, middleware order, route mounting<br>app, cors, database, jwt, security, snmp, queue<br>THE ONLY reader of process.env [INV-02]<br>auth, validate, upload, rateLimit, securityHeaders,<br>errorHandler, auditContext<br>auth, printers, jobs, users, stats, notifications, health<br><br>ipp/ raw9100/ pjl/ select.js<br>client.js oids.js identity.js counters.js<br>pageCounter.js status.js scheduler.js<br>queue.js worker.js convert.js prepare.js dispatch.js<br><br>the single permission gate [INV-01]<br>[INV-07]<br>one file per table, parameterised SQL, no ORM<br>jwt, hash, validators, csv, asyncHandler, logger<br>ordered, forward-only, idempotent | e2e/ |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|

B3. Configuration Reference

Every variable the application reads. Anything not listed here MUST NOT be read from the environment.

| Variable                     | Required | Default     | Notes                                             |
|------------------------------|----------|-------------|---------------------------------------------------|
| NODE_ENV                     | yes      | development | production enables all boot guards                |
| KODE_DEBUG                   | yes      | true        | false in production. Enables the guards in B3.    |
| 3. PORT                      | no       | 3000        | Bound to 127.0.0.1 in production. B18.3           |
| DATABASE_URL<br>credentials. | yes      | none        | Rejected at boot if it contains default<br>GAP-04 |
| DB_POOL_MAX                  | no       | 10          |                                                   |
| JWT_SECRET                   | yes      | none        | Boot fails on placeholder value                   |
| JWT_ACCESS_TTL               | no       | 15m         |                                                   |
| JWT_REFRESH_TTL              | no       | 7d          |                                                   |
| SECRET_KEY                   | yes      | none        | Boot fails on placeholder value                   |
| CORS_ORIGINS<br>production.  | yes      | none        | Explicit allow-list. Wildcard forbidden in        |
| UPLOAD_DIR                   | no       | ./uploads   | Must be outside the web root                      |

| Variable              | Required | Default    | Notes                                               |
|-----------------------|----------|------------|-----------------------------------------------------|
| UPLOAD_MAX_BYTES      | no       | 104857600  | 100 MB                                              |
| UPLOAD_RETENTION_DAYS | no       | 30         | Pending DEC-03. 0 means delete on successful print. |
| LIBREOFFICE_PATH      | no       | autodetect | Health-checked at boot. B10.6                       |
| GHOSTSCRIPT_PATH      | no       | autodetect | Health-checked at boot. B10.6                       |
| CONVERT_TIMEOUT_MS    | no       | 120000     | Per conversion stage. GAP-17                        |
| SNMP_DEFAULT_VERSION  | no       | 2c         |                                                     |
| SNMP_TIMEOUT_MS       | no       | 2000       |                                                     |
| POLL_INTERVAL_MS      | no       | 4000       | Base interval before jitter. B8.2                   |
| SCAN_POLL_INTERVAL_MS | no       | 4000       |                                                     |
| SCAN_STABILITY_MS     | no       | 2000       | File must be unchanged this long. B9.2              |
| QUEUE_CONCURRENCY     | no       | 2          | Deliberately low. B10.4                             |
| QUEUE_MAX_ATTEMPTS    | no       | 3          |                                                     |
| RATE_LIMIT_GENERAL    | no       | 300        | Requests per minute                                 |
| RATE_LIMIT_LOGIN      | no       | 10         | Requests per minute                                 |
| LOG_LEVEL             | no       | info       |                                                     |
| LOG_FORMAT            | no       | json       | B13.1                                               |

### B3.3 Boot guards

With KODE\_DEBUG=false the process MUST refuse to start if any of the following holds. Each failure prints the offending variable name and exits non-zero. A silent start with a bad configuration is worse than no start.

1. JWT\_SECRET or SECRET\_KEY equals a known placeholder
2. DATABASE\_URL contains default credentials such as postgres:postgres (GAP-04, currently unchecked)
3. CORS\_ORIGINS is unset, empty, or \*
4. Any user account still holds the seeded default password (GAP-01)
5. LibreOffice or Ghostscript is not executable at the resolved path (GAP-06)
6. The upload directory is not writable, or resolves inside the served static root

#### B4. Data Model

Target schema. PLANNED marks tables and columns that do not exist in the delivered build. Migrations are forward-only, ordered, and idempotent.

##### B4.1 Conventions

Surrogate integer primary keys. snake\_case . All timestamps TIMESTAMPTZ . Enumerations enforced with CHECK constraints rather than a lookup table, since the sets are small and stable. Every foreign key states its delete behaviour explicitly; there are no implicit cascades.

##### B4.2 zones

Phase 1 does not model buildings, floors or rooms. Printer placement is represented directly by a zone value. A zone is a lightweight classification used for discovery, filtering and reporting. It does not participate in authorization.

Example values are Zone A, Zone B, Zone C and Zone D. They are deployment configuration, not a hard-coded business hierarchy.

##### B4.3 Distributed collectors

No collector table exists in Phase 1. The application server communicates directly with printers. If a future deployment requires a collector because a network segment cannot be reached directly, the collector is introduced as a separate extension and its state is not mixed into the core printer authorization model.

##### B4.4 printers

```
CREATE TABLE printers (
  id SERIAL PRIMARY KEY,
  zone TEXT NOT NULL,
  name TEXT NOT NULL,
  serial_number TEXT,
  identity_anchor TEXT,
  mac_address MACADDR,
  hostname TEXT,
  ip_address INET NOT NULL,
  vendor TEXT,
  model TEXT,
  transport TEXT NOT NULL DEFAULT 'auto'
  CHECK (transport IN ('auto', 'ipp', 'raw9100')),
  ipp_uri TEXT,
  capabilities JSONB NOT NULL DEFAULT '{}'::jsonb,
  capabilities_probed_at TIMESTAMPTZ,
  snmp_version TEXT NOT NULL DEFAULT '2c'
  CHECK (snmp_version IN ('1', '2c', '3', 'disabled')),
  snmp_community TEXT,
  snmp_page_oid TEXT NOT NULL
  DEFAULT '1.3.6.1.2.1.43.10.2.1.4.1.1',
  snmp_print_oid TEXT,
  snmp_copy_oid TEXT,
  last_page_count BIGINT,
  last_page_count_at TIMESTAMPTZ,
  scan_folder TEXT,
  status TEXT NOT NULL DEFAULT 'unknown'
  CHECK (status IN ('online', 'offline', 'degraded', 'unknown')),
  state_reasons TEXT[],
  consecutive_failures INTEGER NOT NULL DEFAULT 0,
  last_checked_at TIMESTAMPTZ,
  is_active BOOLEAN NOT NULL DEFAULT TRUE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE UNIQUE INDEX printers_serial_uq
ON printers (serial_number) WHERE serial_number IS NOT NULL;
CREATE UNIQUE INDEX printers_ip_active_uq
ON printers (ip_address) WHERE is_active;
CREATE INDEX printers_zone_idx ON printers (zone) WHERE is_active;
```

Two deliberate changes from the delivered schema:

- ip\_address is unique only among active printers. A decommissioned device must not block reuse of its address.
- serial\_number is the identity anchor. See B7.1.

## B4.5 users

```

CREATE TABLE users (
  id SERIAL PRIMARY KEY,
  username TEXT NOT NULL UNIQUE,
  email TEXT,
  password_hash TEXT NOT NULL,
  salt TEXT NOT NULL,
  role TEXT NOT NULL CHECK (role IN ('admin','user')),
  department TEXT,
  must_change_password BOOLEAN NOT NULL DEFAULT FALSE, -- PLANNED, GAP-01
  failed_login_count INTEGER NOT NULL DEFAULT 0, -- PLANNED, GAP-18
  locked_until TIMESTAMPTZ, -- PLANNED, GAP-18
  last_login_at TIMESTAMPTZ, -- PLANNED
  is_active BOOLEAN NOT NULL DEFAULT TRUE,
  is_system BOOLEAN NOT NULL DEFAULT FALSE,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  updated_at TIMESTAMPTZ NOT NULL DEFAULT now()
);

```

department is retained for reporting only. It MUST NOT influence access. INV-01 .

A single system account ( is\_system = TRUE , is\_active = FALSE , not loggable-in) owns walk-up jobs so that every job row has a stable, non-impersonatable actor.

## B4.6 user\_printers

```

CREATE TABLE user_printers (
  user_id INTEGER NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  printer_id INTEGER NOT NULL REFERENCES printers(id) ON DELETE CASCADE,
  granted_by INTEGER REFERENCES users(id) ON DELETE SET NULL, -- PLANNED
  granted_at TIMESTAMPTZ NOT NULL DEFAULT now(), -- PLANNED
  PRIMARY KEY (user_id, printer_id)
);

```

The single source of truth for access. INV-01 . Cascade here is correct and intentional: a grant has no meaning once either side is gone, and the audit of who granted it survives in audit\_log .

## B4.7 jobs

```

CREATE TABLE jobs (
  id BIGSERIAL PRIMARY KEY,
  printer_id INTEGER REFERENCES printers(id) ON DELETE RESTRICT, -- INV-05
  user_id INTEGER REFERENCES users(id) ON DELETE SET NULL,
  username_snapshot TEXT NOT NULL, -- INV-06
  printer_name_snapshot TEXT NOT NULL, -- PLANNED, INV-06
  source TEXT NOT NULL
    CHECK (source IN ('app', 'walkup', 'manual')), -- PLANNED
  job_type TEXT NOT NULL
    CHECK (job_type IN ('print', 'copy', 'fax', 'unknown')),
  status TEXT NOT NULL
    CHECK (status IN ('queued', 'processing', 'sent',
                      'completed', 'failed', 'cancelled')),
  pages INTEGER NOT NULL DEFAULT 0,
  copies INTEGER NOT NULL DEFAULT 1,
  impressions INTEGER, -- PLANNED, B8.3
  color_mode TEXT CHECK (color_mode IN ('color', 'grayscale')),
  duplex BOOLEAN,
  document_name TEXT,
  file_path TEXT,
  file_hash TEXT, -- PLANNED, sha256
  print_options JSONB NOT NULL DEFAULT '{}::jsonb', -- PLANNED
  transport_used TEXT, -- PLANNED
  attempts INTEGER NOT NULL DEFAULT 0, -- PLANNED, B10.4
  max_attempts INTEGER NOT NULL DEFAULT 3, -- PLANNED
  next_attempt_at TIMESTAMPTZ NOT NULL DEFAULT now(), -- PLANNED
  locked_by TEXT, -- PLANNED
  locked_at TIMESTAMPTZ, -- PLANNED
  error_code TEXT, -- PLANNED
  notes TEXT,
  created_at TIMESTAMPTZ NOT NULL DEFAULT now(),
  completed_at TIMESTAMPTZ
);

CREATE INDEX jobs_queue_idx
  ON jobs (next_attempt_at) WHERE status = 'queued';
CREATE INDEX jobs_printer_created_idx ON jobs (printer_id, created_at DESC);
CREATE INDEX jobs_user_created_idx ON jobs (user_id, created_at DESC);
CREATE INDEX jobs_created_idx ON jobs (created_at DESC);

```

ON DELETE RESTRICT on printer\_id is the schema-level enforcement of INV-05 . The delivered build blocks hard deletion in the model layer only, which one bypassing route would defeat.

impressions records what the print engine will actually mark ( pages × copies , doubled where the device counts duplex sides separately). This is the quantity the SNMP counter observes, and the ledger in B8.3 depends on it.

## B4.8 notifications PLANNED (GAP-11)

```

CREATE TABLE notifications (
  id          BIGSERIAL PRIMARY KEY,
  type        TEXT          NOT NULL,
  severity    TEXT          NOT NULL CHECK (severity IN ('info','warning','critical')),
  printer_id  INTEGER        REFERENCES printers(id) ON DELETE SET NULL,
  job_id      BIGINT         REFERENCES jobs(id)     ON DELETE SET NULL,
  message     TEXT          NOT NULL,
  payload     JSONB,
  created_at  TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE TABLE notification_reads (
  notification_id BIGINT NOT NULL REFERENCES notifications(id) ON DELETE CASCADE,
  user_id         INTEGER NOT NULL REFERENCES users(id)         ON DELETE CASCADE,
  read_at         TIMESTAMPTZ NOT NULL DEFAULT now(),
  PRIMARY KEY (notification_id, user_id)
);

```

This replaces the in-memory FIFO array, which is lost on restart and has no per-admin read state when more than one admin is watching.

## B4.9 audit\_log PLANNED (GAP-10, DEC-04)

```

CREATE TABLE audit_log (
  id          BIGSERIAL PRIMARY KEY,
  actor_user_id INTEGER REFERENCES users(id) ON DELETE SET NULL,
  actor_username TEXT          NOT NULL,
  action       TEXT          NOT NULL,
  entity_type  TEXT          NOT NULL,
  entity_id    TEXT,
  before       JSONB,
  after        JSONB,
  ip_address   INET,
  created_at   TIMESTAMPTZ NOT NULL DEFAULT now()
);

CREATE INDEX audit_log_created_idx ON audit_log (created_at DESC);
CREATE INDEX audit_log_entity_idx ON audit_log (entity_type, entity_id);

```

Append-only. No UPDATE or DELETE path exists in the application. Written in the same transaction as the change it records ( INV-07 ).

Actions that MUST be audited: user create, update, deactivate, password set; printer create, update, deactivate, delete; printer permission grant and revoke; login success, login failure, logout-all; configuration change through the API.

A permission system with no record of permission changes is not auditable. That is the reasoning behind DEC-04.

B4.10 Quotas and cost accounting – deferred

No quota table is required for Phase 1. The reporting layer records actual application jobs/impressions and observed device activity; it does not enforce spending, page quotas or chargeback. If a future release requires quotas, they can be added without changing the print transport or job lifecycle.

B4.11 refresh\_tokens

B4.11 refresh\_tokens

```
CREATE TABLE refresh_tokens (
  id          BIGSERIAL PRIMARY KEY,
  user_id     INTEGER      NOT NULL REFERENCES users(id) ON DELETE CASCADE,
  family_id   UUID         NOT NULL,                -- PLANNED, B12.3
  token_hash  TEXT         NOT NULL UNIQUE,
  issued_at   TIMESTAMPTZ NOT NULL DEFAULT now(),
  expires_at  TIMESTAMPTZ NOT NULL,
  revoked_at  TIMESTAMPTZ,
  replaced_by BIGINT       REFERENCES refresh_tokens(id) ON DELETE SET NULL,
  user_agent  TEXT,
  ip_address  INET
);
```

family\_id is PLANNED and is what makes reuse detection actionable rather than merely observable. See B12.3.

B5. API Contract

All routes mount under /api . Every route except POST /auth/login , POST /auth/refresh and GET /health requires Authorization: Bearer <access token> .

The authoritative machine-readable contract is an OpenAPI 3.1 document at docs/openapi.yaml , generated from route validators and kept in the repository. It is a release blocker if the OpenAPI document and the implemented routes disagree. The tables below are the architectural inventory, not a substitute for it.

B5.1 Endpoint inventory

| Group | Endpoint           | Role   | Notes                                                |
|-------|--------------------|--------|------------------------------------------------------|
| Auth  | POST /auth/login   | public | Rate limited to 10/min. Returns mustChangePassword . |
|       | POST /auth/refresh | public | Rotates. Reuse revokes the whole family. B12.3       |

| Group         | Endpoint                                                                                      | Role             | Notes                                                |
|---------------|-----------------------------------------------------------------------------------------------|------------------|------------------------------------------------------|
|               | POST /auth/logout                                                                             | any              | Revokes one token                                    |
|               | POST /auth/logout-all                                                                         | any              | Revokes every token for the user                     |
|               | POST /auth/change-password                                                                    | any              | Required path when mustChangePassword                |
|               | GET /auth/me                                                                                  | any              |                                                      |
| Sites         | GET /sites · POST /sites · PUT /sites/:id                                                     | admin for writes | PLANNED                                              |
| Printers      | GET /printers                                                                                 | any              | Non-admins see only permitted printers               |
|               | POST /printers · PUT /:id · DELETE /:id · PUT /:id/active                                     | admin            | Delete conditional, B4.7                             |
|               | POST /printers/:id/check-status                                                               | admin            |                                                      |
|               | POST /printers/:id/probe                                                                      | admin            | PLANNED , capability probe, B7.3                     |
|               | POST /printers/:id/print-file                                                                 | permission-gated | Enqueues, returns 202 with job id                    |
| Jobs          | GET /jobs                                                                                     | any              | Filtered by permission unless admin. Paginated, B5.4 |
|               | GET /jobs/:id                                                                                 | owner or admin   |                                                      |
|               | POST /jobs                                                                                    | admin            | Manual log entry                                     |
|               | POST /jobs/:id/retry                                                                          | owner or admin   | PLANNED                                              |
|               | GET /jobs/export                                                                              | any              | CSV, bounded, B5.4                                   |
| Users         | GET /users · GET /:id · POST / · PUT /:id · PUT /:id/active · PUT /:id/password · DELETE /:id | admin            |                                                      |
|               | PUT /users/:id/printers                                                                       | admin            | The only write path to user_printers . INV-01        |
| Stats         | GET /stats                                                                                    | any              | Scoped to permitted printers unless admin            |
| Notifications | GET /notifications · POST /notifications/:id/read · POST /notifications/read-all              | admin            | PLANNED , persisted, B4.8                            |
| Audit         | GET /audit                                                                                    | admin            | PLANNED , read-only, paginated                       |

| Group      | Endpoint                                                                    | Role              | Notes                     |
|------------|-----------------------------------------------------------------------------|-------------------|---------------------------|
| Uploads    | GET /uploads · GET /uploads/:filename                                       | admin             | Confined directory, B16.4 |
| Collectors | POST /collectors/heartbeat · POST /collectors/events · GET /collectors/work | collector key     | PLANNED , B11.3           |
| Health     | GET /health · GET /health/ready                                             | public / internal | B13.3                     |

### B5.2 Standard error shape

Every non-2xx response body MUST match this shape exactly. No route returns a bare string or an unwrapped driver error. INV-12 .

```
{
  "error": {
    "code": "PRINTER_UNREACHABLE",
    "message": "Printer did not accept the connection.",
    "details": { "printerId": 12, "transport": "ipp" },
    "requestId": "01J9X4K2P8QW3ZB6"
  }
}
```

requestId is generated per request, returned in the X-Request-Id header, and included in every log line for that request. It is the join key between a user's complaint and the logs.

### B5.3 Error taxonomy

| Code                     | HTTP | Meaning                                     |
|--------------------------|------|---------------------------------------------|
| VALIDATION_FAILED        | 400  | Body or query failed schema validation      |
| FILE_TYPE_REJECTED       | 400  | Extension or magic-byte check failed. B10.2 |
| FILE_TOO_LARGE           | 413  | Above UPLOAD_MAX_BYTES                      |
| UNAUTHENTICATED          | 401  | Missing, malformed or expired access token  |
| TOKEN_REUSE_DETECTED     | 401  | Refresh family revoked. B12.3               |
| PASSWORD_CHANGE_REQUIRED | 403  | Only /auth/change-password is permitted     |
| FORBIDDEN                | 403  | Authenticated but not permitted             |
| NOT_FOUND                | 404  |                                             |
| PRINTER_INACTIVE         | 409  | Soft-disabled printer                       |
| PRINTER_HAS_HISTORY      | 409  | Hard delete refused. INV-05                 |
| RATE_LIMITED             | 429  | Includes Retry-After                        |

| Code                | HTTP | Meaning                            |
|---------------------|------|------------------------------------|
| PRINTER_UNREACHABLE | 502  | No transport succeeded             |
| CONVERSION_FAILED   | 502  | Every fallback in B10.3 exhausted  |
| CONVERSION_TIMEOUT  | 504  | Exceeded CONVERT_TIMEOUT_MS        |
| INTERNAL_ERROR      | 500  | Generic. Detail goes to logs only. |

#### B5.4 Pagination and export bounds PLANNED (GAP-19)

GET /jobs and GET /audit MUST be keyset paginated on (created\_at, id) . Default page size 50, maximum 200. Offset pagination is forbidden: it degrades badly once the jobs table holds a year of data.

GET /jobs/export MUST stream the CSV rather than buffering it, and MUST require an explicit date range no wider than 366 days. An unbounded export of the full jobs table is a memory exhaustion vector as well as a slow query.

#### B5.5 Client Print Experience

The client experience is part of the Phase 1 system contract. The goal is that a user can print normally without understanding queues, workers, transports, SNMP or audit records.

Required flow:

Upload -> validate -> prepare -> preview -> choose authorized printer -> choose print options -> confirm  
-> 202 Accepted -> progress -> completed / failed

The UI MUST support copies, colour/grayscale, duplex and page selection. Office files are converted to the final printable representation before page selection is applied so that page numbers correspond to what will actually be printed.

The printer picker MUST show only printers authorized through user\_printers. Zone and recent-printer information may improve ordering, but neither may grant access. The UI SHOULD remember the last successfully used printer.

Printer status is displayed as Available, Warning, Unavailable or Unknown. Unknown MUST NOT be presented as healthy. After submission, the browser polls GET /jobs/:id at a modest interval and translates internal states into simple user-facing progress. Technical error codes remain in the backend; users receive actionable messages. Queue internals, worker IDs, SNMP details, audit terminology and transport implementation details are not exposed to ordinary users.

#### B6. Printer Transport

##### B6.1 Decision

IPP is the primary transport. RAW/9100 is the fallback. ADR-001.

The delivered build framed the choice as RAW/9100 against the Windows print spooler. That is a false binary: it omits IPP, which is a standard, driverless, cross-platform protocol that every AirPrint or Mopria certified device supports natively, and which returns the feedback RAW/9100 structurally cannot.

What IPP provides that RAW/9100 does not:

| Capability                              | RAW/9100                     | IPP                           |
|-----------------------------------------|------------------------------|-------------------------------|
| Job accepted confirmation               | TCP accepted the bytes       | job-state from the device     |
| Job completed / aborted                 | unknown                      | job-state , job-state-reasons |
| Pages actually marked                   | unknown                      | job-impressions-completed     |
| Printer state (toner, paper, jam, door) | unknown                      | printer-state-reasons         |
| Copies, duplex, colour, media           | PJL, honoured inconsistently | Standard job attributes       |
| Supported options per model             | unknown                      | Get-Printer-Attributes        |
| Native PDF acceptance                   | device dependent             | Mandatory for IPP Everywhere  |

Two of these close known limitations outright: GAP-25 (status is reachability only) and GAP-24 (PJL option support unverified per model). A third removes work: on IPP Everywhere devices the PDF to PostScript conversion stage becomes unnecessary, dropping a Ghostscript dependency from the hot path.

RAW/9100 is retained, not replaced. Older devices and anything that fails the IPP probe continue to work exactly as they do today.

## B6.2 Transport selection PLANNED (GAP-23)

Resolved per printer, cached in printers.transport and printers.capabilities , re-probed when capabilities\_probed\_at is older than 30 days or an admin triggers POST /printers/:id/probe .

```
selectTransport(printer):
  if printer.transport == 'ipp'          -> IPP      (operator override)
  if printer.transport == 'raw9100'      -> RAW      (operator override)

  # transport == 'auto'
  if printer.capabilities.ipp.supported is true -> IPP
  if printer.capabilities.ipp.supported is false -> RAW
  # unknown: probe once, persist the result, then decide
  probe(printer); retry selectTransport(printer)
```

Send with IPP, falling back on failure:

```
send(printer, document, options):
  transport = selectTransport(printer)

  if transport == IPP:
    r = ippPrintJob(printer.ipp_uri, document, mapOptions(options))
    if r.accepted -> return { ok, transport: 'ipp', jobUri: r.jobUri }
    if r.error is transient -> return { retryable } # queue retries, B10.4
    if r.error is protocol -> mark ipp.supported = false # fall through to RAW
  return sendRaw9100(printer, document, options)
```

A protocol-level IPP failure demotes the printer to RAW permanently until the next probe. A transient failure (timeout, connection refused, server-error-busy ) MUST NOT demote it: a printer that is merely asleep is not a printer without IPP. Conflating the two is the most likely defect in this section.

## B6.3 IPP specifics

- Target ipp://<host>:631/ipp/print by default. Store the resolved URI in printers.ipp\_uri ; some vendors use a different path.
- Operation Print-Job (0x0002) for the send, Get-Printer-Attributes (0x000B) for the probe, Get-Job-Attributes (0x0009) for completion polling.
- Prefer application/pdf where document-format-supported includes it. Fall back to application/postscript , then application/octet-stream .
- Option mapping: copies , sides ( one-sided , two-sided-long-edge , two-sided-short-edge ), print-color-mode ( color , monochrome ), media , orientation-requested , page-ranges .
- Use a maintained IPP client library. Hand-rolling a second binary protocol encoder repeats the mistake identified in ADR-005.

## B6.4 RAW/9100 specifics BUILT

Plain TCP to port 9100, bytes written directly, proper half-close (write, flush, FIN), fully asynchronous so one dead printer never blocks another user's request.

A hand-built PJL header and footer carries copies, duplex, orientation, colour mode and paper size. Automatic multi-framing fallback tries, in order: PJL with an explicit LANGUAGE line, PJL with no language line, then the raw file with no PJL at all, keeping the first framing the printer's connection accepts. This was added because some Xerox devices reject one framing and accept another, and it requires no per-printer setup.

PARTIAL. PJL is honoured by most HP and PCL-family devices but is not a universal standard. A printer that ignores it still prints, because RAW-only devices discard the PJL lines as text, but copies, duplex and colour silently do not apply. There is no way to detect this over RAW. This is precisely the limitation IPP removes, and it is why RAW is the fallback rather than the default.

## B7. Printer Identity, Capability and State

### B7.1 Identity PLANNED (GAP-13)

A printer is identified by its serial number, not its IP address. ADR-006.

The delivered schema makes ip\_address unique and treats it as identity. In a DHCP environment across multiple buildings this fails in a specific and damaging way: the device takes a new lease, every poll fails, status flips to offline, and an admin "fixes" it by adding the printer again. The result is two rows, a split audit trail, and a permission set attached to the wrong one.

Target behaviour:

1. On every successful SNMP poll, read prtGeneralSerialNumber ( 1.3.6.1.2.1.43.5.1.1.17.1 ) and sysName ( 1.3.6.1.2.1.1.5.0 ).
2. If the serial does not match the stored value, the device at that address MUST NOT be treated as this printer. Raise a critical notification and mark the printer degraded . Do not silently rebind: a swapped device is exactly the case where silently continuing corrupts the record.
3. If a poll fails but a stored serial exists, an admin MAY re-point the record at a new IP through PUT / printers/:id , preserving the history.
4. Operationally, printers SHOULD hold DHCP reservations. The application must not depend on it, but the estate is far more stable with it.

### B7.2 Adding a printer

An admin supplies IP address, name and site. The system probes and fills in serial, vendor, model, MAC, IPP support and capabilities. The PRD requirement of "add by IP only" is preserved; everything else is discovered rather than typed.

### B7.3 Capability detection PLANNED (GAP-24)

Probe order, results persisted to printers.capabilities :

1. IPP Get-Printer-Attributes where reachable. Read document-format-supported , sides-supported , print-color-mode-supported , copies-supported , media-supported , printer-make-and-model , ipp-versions-supported .
2. SNMP for identity and, where IPP is absent, hrDeviceDescr and prtMarkerColorantValue to infer colour capability.

3. Unknown where neither answers. The UI MUST then present print options as unverified rather than implying they will take effect. Showing a duplex toggle that silently does nothing is worse than not showing it.

```
{
  "ipp": { "supported": true, "versions": ["1.1", "2.0"], "uri": "ipp://10.20.3.14:631/ipp/
print" },
  "formats": ["application/pdf", "application/postscript"],
  "sides": ["one-sided", "two-sided-long-edge"],
  "colorModes": ["color", "monochrome"],
  "maxCopies": 999,
  "media": ["iso_a4_210x297mm", "iso_a3_297x420mm"],
  "probedVia": "ipp",
  "counters": { "life": true, "print": false, "copy": false }
}
```

#### B7.4 State PARTIAL to PLANNED (GAP-25)

Today status is an ICMP ping spawned per printer per sweep. It answers "the network interface responds", not "this device can print right now", and it costs an OS process per check.

Target, in preference order:

1. IPP printer-state and printer-state-reasons where supported. Maps directly to online , degraded ( toner-low , media-low , marker-supply-low ) and offline ( media-empty , jam , door-open , toner-empty , shutdown ), stored in printers.state\_reasons .
2. SNMP hrPrinterDetectedErrorState ( 1.3.6.1.2.1.25.3.5.1.2.1 ) where IPP is absent. A bit field covering low paper, no paper, low toner, no toner, door open, jammed, offline and service requested.
3. TCP connect to 631 or 9100 as the reachability floor. This replaces ICMP: it is cheaper than spawning a process, and it tests the port that actually matters.

Printers that fail a sweep back off exponentially using consecutive\_failures : 4s, 8s, 16s, up to 5 minutes, with jitter. One unreachable building must not consume the poll budget for the rest of the fleet.

#### B8. Walk-Up Job Detection

The requirement (PRD 6.4) is to detect and log jobs started at the device, outside this system. This is the least obvious part of the design and the part most likely to be implemented incorrectly, so it is specified in full.

##### B8.1 Mechanism

The backend polls each printer's Printer-MIB page counter over SNMP. An increase that this system did not cause is a walk-up event.

BUILT : polling every four seconds using prtMarkerLifeCount ( 1.3.6.1.2.1.43.10.2.1.4.1.1 , overridable per printer), with a hand-implemented SNMPv1 client (raw BER/ASN.1 encoding, no external package).

PLANNED (GAP-26, ADR-005): replace the hand-rolled client with a maintained library.

The stated reason for hand-rolling was avoiding a dependency. That is the wrong trade here. The code in question is a binary parser consuming unauthenticated UDP from the network, which makes it a security

surface, and it cannot do SNMPv3. A maintained library removes both problems. The existing OID and polling logic port across unchanged; only the wire layer is replaced.

Transport policy: prefer SNMPv2c ( GetBulk , better error reporting), support v3 with authPriv where the fleet allows it, retain v1 for devices that support nothing else. Community strings are secrets ( INV-08 ).

## B8.2 Scheduling

Base interval POLL\_INTERVAL\_MS with per-printer jitter of plus or minus 25 percent, so 50 devices do not fire simultaneously every four seconds. Failed printers back off as in B7.4. Polling never runs concurrently against the same printer.

## B8.3 Attribution: the outstanding-impression ledger PLANNED (GAP-09)

PARTIAL today: immediately before an app-initiated print reaches the socket, the printer's IP is marked with a timestamp. Any counter increase within a fixed 45 second window is treated as "this was us" and skipped.

Two failure modes follow from a fixed window:

- A large document that takes longer than 45 seconds to convert and transmit can have its own counter increase land outside the window and be logged as a walk-up job that never happened.
- Two people printing to one device inside a single poll cycle produce one combined entry with a summed page count, and neither is attributable.

Target design replaces the time window with a ledger of outstanding impressions per printer:

```
onAppJobSent(printer, job):
    ledger[printer.id].push({
        jobId:      job.id,
        impressions: job.impressions,          # pages x copies, doubled if duplex counts sides
        expiresAt:  now + 60s + (job.impressions * 2s), capped at 15 minutes
    })

onCounterDelta(printer, delta):
    if delta < 0:
        # reboot, firmware reset, counter rollover
        resetBaseline(printer, current); notify('counter_reset', warning); return

    expire(ledger[printer.id])                # drop entries past expiresAt

    for entry in ledger[printer.id] oldest first:
        consumed = min(delta, entry.impressions)
        entry.impressions -= consumed
        delta -= consumed
        if entry.impressions == 0: remove(entry)
        if delta == 0: return

    if delta > 0:
        logWalkUpJob(printer, impressions = delta)
```

Why this is better: it reconciles against quantity, not elapsed time. A slow job is still matched whenever its impressions finally appear. A job that genuinely was a walk-up is still logged even if it happened while an app job was in flight, because only the app job's own impression count is absorbed. The time cap exists solely to stop a failed job from absorbing a later real one.

Residual limitation, and it is unavoidable at this layer: if two people print at the same device inside one poll cycle, SNMP returns one number. Individual attribution is impossible from a counter. IPP job attributes (B6.3) solve this for app jobs; for two simultaneous walk-up jobs it remains one combined entry, and that MUST be stated in the report footer rather than left implied.

#### B8.4 The measurement flaw: copies and faxes PLANNED (GAP-08)

prtMarkerLifeCount counts every impression the print engine marks. That includes photocopies, received faxes and internal report pages. The delivered system logs photocopies as walk-up prints.

On an MFP fleet in a sports club this is material, not marginal. Required handling:

1. Where vendor counters exist, store the print-only and copy-only OIDs in printers.snmp\_print\_oid and printers.snmp\_copy\_oid , poll them alongside the life counter, and classify the delta into job\_type print , copy or fax . The inventory pass (B19.2) captures these per model; they differ across HP, Xerox, Canon and Konica Minolta and there is no standard equivalent.
2. Where they do not exist, the job is logged with job\_type = 'unknown' and source = 'walkup' . It MUST NOT be recorded as print .
3. Reports MUST NOT aggregate unknown into a figure labelled "prints". Label it device activity. DEC-06.

#### B8.5 Silent gaps

GAP and unavoidable: a printer with SNMP disabled, or blocked by a firewall or VLAN rule, reports no walk-up activity at all. From an admin's point of view this is indistinguishable from a printer nobody walks up to.

Required mitigation, since the gap itself cannot be closed: printers.snmp\_version = 'disabled' is set explicitly during inventory, the dashboard MUST show which printers have walk-up tracking unavailable, and any report covering those printers MUST carry a coverage note. A known gap that is visible is a limitation. A known gap that is invisible is a false report.

#### B8.6 Spoofing

PARTIAL . SNMP replies are accepted only from the exact address queried, since UDP has no sender authentication. This prevents a spoofed reply injecting a fabricated walk-up entry from an arbitrary host. It does not add authentication that SNMPv1 lacks. SNMPv3 with authPriv is the real fix where the fleet supports it (B8.1).

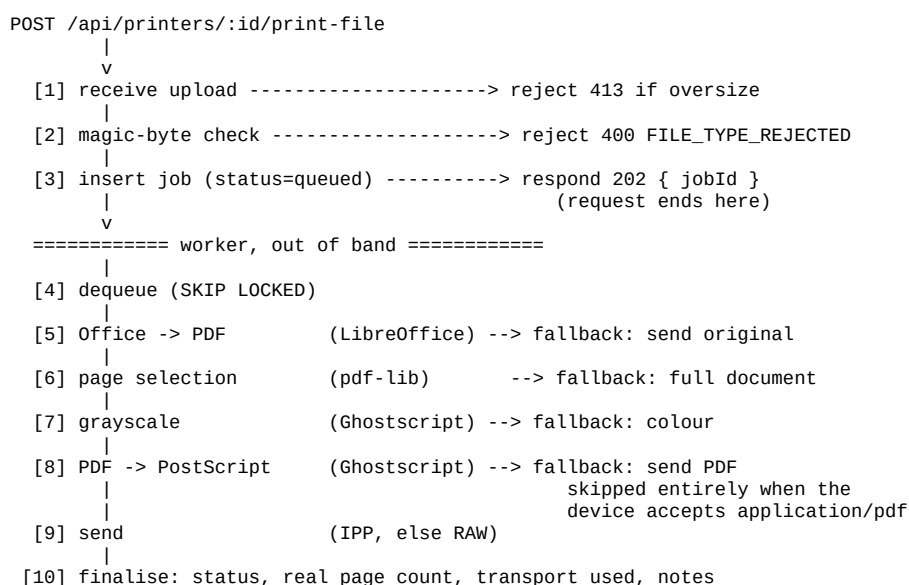
### B9. Scan Detection – OUT OF SCOPE FOR PHASE 1

The previous architecture included scan-to-folder ingestion and file-stability handling. Those mechanisms are not part of the Phase 1 printer application because the product requirement is printing, printer monitoring and print activity accounting. Scan management may be added later as an independent capability.

No scan watcher, scan share, scan job type, scan ingestion API or scan-specific retention policy is required for Phase 1.

### B10. Print Pipeline and Job Queue

#### B10.1 Flow



Step 3 is a change from the delivered build, which handles the whole pipeline inline in the HTTP request. See B10.4.

#### B10.2 Upload safety BUILT

Extensions are whitelisted at the multer layer ( pdf, txt, doc, docx, xls, xlsx, png, jpg, jpeg ), then re-checked against actual file content: leading bytes are matched against what the extension claims (PDF, PNG, JPEG, legacy OLE, OOXML zip), and anything carrying an executable or script signature ( MZ , ELF , #! ) is rejected outright regardless of extension. Size is capped at UPLOAD\_MAX\_BYTES . Files are stored under a random, collision-free name.

INV-09 : no byte reaches a converter or a printer before this passes.

#### B10.3 Conversion PARTIAL to PLANNED

Every stage is best-effort: a failed stage falls back to the previous stage's output rather than failing the job. That behaviour is correct and is retained.

Three defects to fix:

GAP-16, LibreOffice concurrency. Concurrent headless soffice invocations collide on a shared user profile lock; the second silently hangs or fails. Every invocation MUST pass its own profile directory:

```
soffice --headless --norestore \
  --env:UserInstallation=file:///tmp/lo-<jobId> \
  --convert-to pdf --outdir <out> <in>
```

The profile directory is removed after the run, in a finally block.

GAP-17, no timeouts. A malformed document can hang a converter indefinitely, and with no queue that consumes a request slot forever. Every converter invocation MUST run under CONVERT\_TIMEOUT\_MS , be killed on expiry ( SIGKILL after SIGTERM ), and surface CONVERSION\_TIMEOUT .

GAP-20, no isolation. See B16.2.

Page counts come from pdf-lib after page selection. The byte-scan fallback is unreliable and any count derived from it MUST be flagged in jobs.notes , because these numbers feed reports.

#### B10.4 The queue PLANNED (GAP-12)

Volume is low: a handful of people printing. The reason to queue is not throughput.

The reasons are: a job survives a server restart instead of vanishing; a printer that is mid-reboot gets an automatic retry instead of a failure the user must notice and repeat; a 60 second Office conversion does not sit inside an HTTP request where a browser or proxy timeout kills it and leaves the user with no idea whether it printed; and LibreOffice and Ghostscript, being single-process, serialise anyway, so the only question is whether they serialise gracefully or by blocking the event loop.

PostgreSQL, not Redis. ADR-004. Postgres is already a hard dependency and already the place job state must be durable for audit reasons. Redis and BullMQ would add a second stateful service, a second backup concern and a second failure mode to a single on-premise server, in exchange for throughput this system will never need.

Dequeue:

```
UPDATE jobs
  SET status      = 'processing',
      locked_by   = $1,
      locked_at   = now(),
      attempts    = attempts + 1
 WHERE id = (
   SELECT id FROM jobs
   WHERE status = 'queued'
   AND next_attempt_at <= now()
   ORDER BY created_at
   FOR UPDATE SKIP LOCKED
   LIMIT 1
 )
RETURNING *;
```

- QUEUE\_CONCURRENCY defaults to 2, deliberately low, because the converters are the bottleneck.
- Retry on transient failures only (connection refused, timeout, server-error-busy ). Backoff 30s, 2m, 10m. Permanent failures ( FILE\_TYPE\_REJECTED , PRINTER\_INACTIVE , authentication) MUST NOT retry.
- On startup, any job left processing with locked\_at older than 15 minutes is reset to queued . This is what makes a crash recoverable.
- After max\_attempts , the job is failed with error\_code set and a warning notification raised.

#### B10.5 Duplicate submission

A job is de-duplicated on (user\_id, printer\_id, file\_hash) within 60 seconds and rejected with a clear message. Users double-click, and printing a second copy is not a harmless mistake.

#### B10.6 Converter prerequisites GAP-06

LibreOffice and Ghostscript are OS-level binaries, not npm packages. The delivered code degrades gracefully when they are absent, which is the correct runtime behaviour and a dangerous deployment behaviour: a fresh host without them appears to work while silently skipping every Office conversion and every grayscale request.

At boot, the application MUST execute each binary with a version flag, log the resolved path and version, and, when KODE\_DEBUG=false , refuse to start if either is missing. Graceful degradation is correct mid-flight and wrong at startup.

#### B10.7 Retention DEC-03

Uploaded originals are retained UPLOAD\_RETENTION\_DAYS (recommended 30), then purged by a scheduled sweep. The jobs row and its metadata are retained indefinitely; only the file bytes are removed. Purge activity is logged. Disk usage on the upload volume is monitored and alerted (B13.4).

### B11. Distributed Network Extension – DEFERRED

A collector is not deployed, configured or required in Phase 1. The core application communicates directly with configured printers. There is no collector table, heartbeat API, offline spool, collector credential or collector monitoring requirement in the Phase 1 data model.

If a later deployment proves that a printer segment is not directly reachable and policy prevents the required inbound printer protocols, a collector may be introduced as a separate deployment extension. The extension MUST preserve the existing central authorization, job and reporting model.

### B12. Authentication and Authorisation

#### B12.1 Roles

Two roles: admin and user . Deliberately flat. A third role has been considered and rejected: at this size it adds a permission matrix to maintain without a real separation of duty behind it.

## B12.2 Permission matrix

| Capability                               | user                       | admin |
|------------------------------------------|----------------------------|-------|
| Log in, change own password              | yes                        | yes   |
| List printers                            | permitted only             | all   |
| Print to a printer                       | if in user_printers        | any   |
| View jobs                                | own, on permitted printers | all   |
| Export report                            | own scope                  | all   |
| Retry own failed job                     | yes                        | yes   |
| Create, edit, disable printers and sites | no                         | yes   |
| Grant or revoke printer access           | no                         | yes   |
| Manage users                             | no                         | yes   |
| View notifications and audit log         | no                         | yes   |
| Browse uploads                           | no                         | yes   |

INV-01 : the "if in user\_printers " cell is evaluated by services/printerAccess.js and nowhere else.

## B12.3 Tokens BUILT , one addition PLANNED

Passwords are hashed and salted, never stored or logged in plain text. New deployments SHOULD use argon2id.

Access token: short-lived JWT, default 15 minutes, Authorization: Bearer .

Refresh token: opaque, single-use, rotating, stored hashed, default 7 day expiry. POST /auth/refresh issues a new pair and revokes the old, chained through replaced\_by so reuse of a stolen token is detectable.

PLANNED : detection needs a response. Detecting reuse and doing nothing about it provides no security. Every refresh token carries a family\_id (B4.11). When a token that is already revoked\_at is presented, the entire family MUST be revoked immediately, the session terminated, TOKEN\_REUSE\_DETECTED returned, and a critical notification and audit entry written. Reuse of a rotated token means either a stolen token or a broken client; both warrant ending the session.

/logout revokes one token. /logout-all revokes every token for the user.

## B12.4 Account security gaps

GAP-01, blocking. A default admin account ( admin / admin123 ) is seeded on first boot when the users table is empty, with no forced rotation. Required: the seeded account is created with must\_change\_password = TRUE ; while that flag is set every route except /auth/change-password returns PASSWORD\_CHANGE\_REQUIRED ; and with KODE\_DEBUG=false the process refuses to boot if any account still holds the seeded password. An operational reminder is not a control.

GAP-18. No password complexity policy, no account lockout, no password reset flow. Rate limiting throttles by IP, not by account, so it does not stop a slow distributed attempt against one username. Required: minimum length 12 with a breached-password denylist; lockout via failed\_login\_count and locked\_until after 10 failures in 15 minutes, per account and per IP; admin-triggered forced reset that sets must\_change\_password .

GAP-29, v2.0. No MFA. Acceptable for an internal system behind the club network. It should be a conscious acceptance, not an oversight, and it is recorded here as one.

B13. Observability PLANNED (GAP-21)

Currently there is no structured logging, no metrics and no alerting. For an on-premise single-server system with no operator watching it, that is the difference between a fault noticed in minutes and one noticed in a quarter.

B13.1 Logging

Structured JSON to stdout ( pino or equivalent), collected by the process manager. Every line carries timestamp , level , requestId , userId where authenticated, printerId where relevant, jobId where relevant, event , durationMs .

Never logged, INV-08 : passwords, tokens, SNMP community strings, collector API keys, uploaded file contents.

Log rotation is the process manager's responsibility and is a go-live blocker (GAP-03).

B13.2 Metrics

Prometheus-format /metrics , bound to localhost. Minimum set:

```
kode_print_jobs_total{status,transport} · kode_print_job_duration_seconds{stage} ·
kode_queue_depth · kode_queue_oldest_seconds · kode_printers_up ·
kode_printer_poll_failures_total{printer} · kode_walkup_events_total{type} ·
kode_conversion_failures_total{stage} · kode_auth_failures_total · kode_upload_dir_bytes
```

B13.3 Health endpoints

GET /health returns 200 whenever the process is alive. Used by the process manager.

GET /health/ready checks database connectivity, converter binaries, upload directory writability and queue worker liveness, returning per-check status. Used by monitoring and by the go-live checklist.

B13.4 Alerting

KODE already runs Zabbix. Use it rather than introducing a second monitoring stack.

| Alert        | Condition                        | Severity |
|--------------|----------------------------------|----------|
| Service down | /health fails twice in 60s       | critical |
| Not ready    | /health/ready degraded for 5 min | high     |

| Alert                       | Condition                                  | Severity                                      |
|-----------------------------|--------------------------------------------|-----------------------------------------------|
| Queue stalled               | kode_queue_oldest_seconds > 900            | high                                          |
| Printer down                | one printer unreachable for 15 min         | warning                                       |
| Fleet down<br>not a printer | more than 30 percent unreachable for 5 min | critical, indicates a network fault,<br>fault |
| Disk                        | upload volume above 80 percent             | warning                                       |
| Auth attack                 | kode_auth_failures_total above 50 in 5 min | warning                                       |
| Collector silent            | no heartbeat for 3 intervals               | critical                                      |
| Backup                      | nightly dump missing or non-zero exit      | high                                          |

#### B14. Failure Modes

Behaviour under each named fault. Anything not listed here is undefined behaviour and a specification defect.

| Fault                                            | Detection                       | Required behaviour                                                                                |
|--------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------------------|
| Printer powered off mid-job                      | Send timeout                    | Retry per B10.4, then failed with PRINTER_UNREACHABLE and a notification                          |
| 1) Printer IP changed                            | Serial mismatch or poll failure | degraded , critical notification, no silent rebind (B7.                                           |
| Printer replaced with a<br>action different unit | Serial mismatch                 | Refuse to bind, critical notification, require admin                                              |
| SNMP disabled or blocked                         | No reply                        | snmp_version = 'disabled' , walk-up tracking flagged unavailable, coverage note on reports (B8.5) |
| Counter decreases                                | Negative delta                  | Reset baseline, warning , log no job (B8.3)                                                       |
| LibreOffice missing<br>note in                   | Boot health check               | Refuse to boot in production; degrade with a flagged development (B10.6)                          |
| Conversion hangs                                 | CONVERT_TIMEOUT_MS              | Kill process, CONVERSION_TIMEOUT , retry once, then fail                                          |
| Database unreachable                             | Pool error                      | 503 on all routes, /health/ready fails, worker pauses without consuming attempts                  |
| Server restarts mid-job                          | locked_at staleness             | Jobs in processing return to queued (B10.4)                                                       |
| Disk full on upload<br>jobs,<br>volume           | Write error                     | Reject uploads with a clear error, keep printing queued<br>critical alert                         |
| Scan share<br>recorded<br>unreachable            | Watcher error                   | warning , retry with backoff, no job logged, gap                                                  |

| Fault                                             | Detection          | Required behaviour                                          |
|---------------------------------------------------|--------------------|-------------------------------------------------------------|
| Two walk-up jobs in reports (B8.3) one poll cycle | Single delta       | One combined entry, stated limitation, footnote in          |
| Refresh token reused                              | revoked_at present | Revoke family, end session, critical notification (B12.3)   |
| Collector uplink lost critical                    | Missed heartbeats  | Collector spools locally and replays; server raises (B11.4) |

#### B15. Non-Functional Requirements

Targets, not aspirations. Each is measurable and each has a stated basis.

| ID             | Requirement                                      | Target              | Basis                                            |
|----------------|--------------------------------------------------|---------------------|--------------------------------------------------|
| NFR-01         | API response time, non-print routes              | p95 under 300 ms    | Fleet of 50 rows, indexed queries                |
| NFR-02         | Time from submit to accepted ( 202 )             | p95 under 2 s       | Upload plus magic-byte check only                |
| NFR-03         | Time from submit to bytes at the printer, PDF    | p95 under 15 s      | No conversion required                           |
| NFR-04         | Time from submit to bytes at the printer, Office | p95 under 90 s      | LibreOffice single-process                       |
| NFR-05         | Concurrent users supported                       | 15 concurrent users | Phase 1 target with headroom                     |
| NFR-06         | Fleet size supported                             | 12 printers         | Phase 1 deployment profile                       |
| NFR-07         | Walk-up detection latency                        | under 10 s          | Four second poll plus jitter                     |
| NFR-08         | Scan detection latency                           | under 15 s          | Poll plus stability gate                         |
| NFR-09         | Availability, business hours                     | 99 percent          | Single server, supervised restart                |
| NFR-10         | Recovery time objective                          | 4 hours             | Restore from nightly dump, B19.4                 |
| NFR-11<br>club | Recovery point objective                         | 24 hours            | Nightly backup. Tighten only if the requires it. |
| NFR-12<br>not  | Job record retention                             | 3 years<br>minimum  | Audit purpose. DEC-03 governs files, records.    |
| NFR-13         | Queue drain after 1 hour outage                  | under 15 min        | Concurrency 2 against expected volume            |

Scale beyond 50 printers or 30 concurrent users invalidates NFR-01 through NFR-04 and requires re-testing, not re-assumption.

B16. Threat Model

Three inputs arrive from outside the trust boundary and each is treated as hostile.

B16.1 Untrusted input 1: uploaded documents

PARTIAL . Extension whitelist plus magic-byte verification (B10.2) confirms a file is what it claims. It does not detect an exploit inside a well-formed document. A genuine PDF or DOCX can carry a payload targeting the parser, and both LibreOffice and Ghostscript are large C++ parsers with a long history of parser vulnerabilities.

PLANNED (GAP-20). Antivirus is the weaker control here and is not the primary mitigation. Process isolation is:

- Converters run as a dedicated unprivileged account with no network access and no write access outside a per-job temporary directory
- Hard wall-clock timeout, memory limit and process count limit per invocation
- Containerised where the host supports Docker; a restricted service account where it does not
- LibreOffice invoked with --norestore and a per-job profile (B10.3), so a crash cannot carry state into the next job
- Optional ClamAV scan before conversion where the club's policy requires it, as defence in depth rather than as the control

B16.2 Untrusted input 2: SNMP replies

PARTIAL . Replies are accepted only from the queried address (B8.6). The hand-rolled BER parser (GAP-26) consumes attacker-influenceable bytes and is the single least reviewed piece of security-relevant code in the system. Replacing it with a maintained library is a security fix, not a refactor.

B16.3 Untrusted input 3: scan share files

Filenames from printer firmware reach the filesystem and the database. All filenames are sanitised, length-capped, and never interpolated into a path ( INV-04 , B16.4). File contents are stored and served, never parsed.

B16.4 OWASP Top 10 coverage

| Category                      | Status  | Control                                                                                      |
|-------------------------------|---------|----------------------------------------------------------------------------------------------|
| A01 Broken access control     | BUILT   | Single permission gate, INV-01, permission matrix B12.2                                      |
| A02 Cryptographic failures    | PARTIAL | Hashed passwords and tokens. TLS is a deployment step, GAP-02.                               |
| A03 Injection                 | BUILT   | Parameterised SQL everywhere, INV-04. No shell string concatenation in converter invocation. |
| A04 Insecure design code.     | PARTIAL | This document is the mitigation. Threat model maintained alongside the                       |
| A05 Security misconfiguration | PARTIAL | Boot guards B3.3. GAP-01 and GAP-04 open.                                                    |

| Category                  | Status  | Control                                                                                           |
|---------------------------|---------|---------------------------------------------------------------------------------------------------|
| A06 Vulnerable components | GAP     | No dependency audit in CI. B17.4                                                                  |
| A07 Auth failures         | PARTIAL | Rotation and rate limiting built. Lockout, reset and complexity are GAP-18.                       |
| A08 Integrity failures    | PARTIAL | Magic-byte checks. No signed artifacts or SBOM.                                                   |
| A09 Logging failures      | GAP     | GAP-10 audit log, GAP-21 structured logging                                                       |
| A10 SSRF user-            | BUILT   | Printer addresses are admin-supplied and restricted to private ranges; no supplied URL is fetched |

B17. Test Strategy PLANNED (GAP-22)

The delivered artifact contains no automated tests. For a system whose output is an audit record, that is the highest-leverage gap in v1.1: nothing else on the list protects the numbers.

B17.1 Layers

Unit. Pure logic with no I/O: the impression ledger (B8.3), transport selection (B6.2), PJL construction, permission evaluation, magic-byte detection, retry and backoff calculation. Jest.

Integration. Routes against a real PostgreSQL instance in a container, with printers stubbed at the socket layer. Every endpoint in B5.1, both authorised and unauthorised. Jest and Supertest.

Protocol. A fake printer harness: a TCP listener on 9100 that records what it receives, an IPP responder returning canned attribute sets, and an SNMP agent serving a scriptable counter. This is what makes the hard parts testable without hardware, and it is the piece most likely to be skipped and most costly to skip.

End to end. Playwright over the critical journeys: log in, upload, select options, print, see the job logged; admin grants access and the user sees the printer; walk-up event appears in the report.

B17.2 Mandatory scenarios

These are the cases where a defect corrupts the record rather than annoying a user. Each MUST have a named test.

1. Counter increases by exactly the app job's impressions: no walk-up logged
2. Counter increases by more than the app job's impressions: only the excess is logged
3. Counter increases with no app job outstanding: full delta logged as walk-up
4. Counter decreases: baseline reset, no job logged
5. App job that takes longer than the legacy 45 second window: still correctly attributed
6. Two app jobs outstanding on one printer, deltas arriving out of order
7. Hard-deleting a printer with job history: refused by the database, not only by the model
8. Deleting a user: jobs retain a readable username via the snapshot
9. Reusing a rotated refresh token: whole family revoked
10. Non-admin printing to a printer not in user\_printers : 403, no job row created

- 11. Server killed mid-job: job returns to queued and completes on restart
- 12. Executable renamed to .pdf : rejected before reaching any converter
- 13. Conversion exceeding the timeout: killed, job fails cleanly, worker stays alive
- 14. Partially written scan file: not ingested until stable
- 15. IPP transient failure: retried, printer not demoted to RAW

B17.3 Traceability

Each PRD requirement in A7 maps to a code path and at least one named test. The matrix is maintained in test/TRACEABILITY.md and reviewed at each release. A requirement with no test is not delivered, regardless of whether it demonstrates.

B17.4 CI

GitHub Actions, dev and main branches per the department pipeline.

Pull request into dev : lint, typecheck, unit and integration tests, npm audit at moderate and above, licence check. Merge into main : full suite including end to end, build, deploy artifact.

Coverage targets apply to the pipeline, transport, permission and ledger modules, not as a blanket percentage across the repository.

B18. Deployment Topology

B18.1 Target host

Company-managed Windows VM on the internal network, consistent with KODE Technology's standard deployment. Docker Compose where the host supports it, otherwise PM2 under a Windows service with IIS as reverse proxy and TLS terminator.

Minimum specification: 4 vCPU, 8 GB RAM, 100 GB disk. The converters, not the Node process, drive the CPU and memory requirement, and disk is sized for uploads plus database plus 30 days of logs.

B18.2 Ports and firewall matrix

| Source            | Destination | Protocol | Port | Direction | Purpose                 |
|-------------------|-------------|----------|------|-----------|-------------------------|
| Staff workstation | Server      | TCP      | 443  | inbound   | Web interface           |
| Server            | Printer     | TCP      | 631  | outbound  | IPP print and probe     |
| Server            | Printer     | TCP      | 9100 | outbound  | RAW fallback            |
| Server            | Printer     | UDP      | 161  | outbound  | SNMP polling            |
| Server            | PostgreSQL  | TCP      | 5432 | localhost | Database, never exposed |

| Source | Destination | Protocol | Port | Direction | Purpose |
|--------|-------------|----------|------|-----------|---------|
|--------|-------------|----------|------|-----------|---------|

B18.3 TLS GAP-02 , blocking

The application does not terminate TLS and MUST NOT be exposed directly. IIS terminates TLS and proxies to Node on 127.0.0.1 . In production the Node process binds to loopback only; binding to 0.0.0.0 with KODE\_DEBUG=false is a boot failure.

Certificate from the club's internal CA or a public certificate for the internal hostname. helmet() remains responsible for security headers behind the proxy, and trust proxy must be set so rate limiting sees the real client address rather than the proxy's.

B18.4 Process supervision GAP-03 , blocking

PM2 ecosystem file with auto-restart, restart on boot registered as a Windows service, memory ceiling, and log rotation with 30 day retention. A crashed Node process that nobody restarts is the most likely cause of an outage in this system.

B19. Operations Runbook

B19.1 Go-live checklist

Every item verified and initialled before the system is announced. This list is exactly A9.1.

- 1. Default admin password rotated; must\_change\_password behaviour verified
- 2. JWT\_SECRET , SECRET\_KEY and database credentials rotated from defaults; boot guards confirmed to fail on placeholders
- 3. TLS terminating at IIS; plain HTTP redirects; Node bound to loopback
- 4. PM2 service registered, auto-restart verified by killing the process, log rotation confirmed
- 5. LibreOffice and Ghostscript verified via /health/ready
- 6. Leftover test uploads removed from the uploads directory (GAP-05)
- 7. Nightly backup running, and a restore tested into a scratch database (an untested backup is not a backup)
- 8. Per-printer inventory complete (B19.2)
- 9. Basic health/readiness and critical alerting verified per B13
- 10. CORS\_ORIGINS set to the real hostname, not a wildcard
- 11. At least one admin account other than the seeded one exists

B19.2 Per-printer inventory

Run once against every device, recorded in the printers table and in a sheet handed to IT. This turns abstract gaps into a concrete list of which devices are limited and how.

| Field                            | How obtained               |
|----------------------------------|----------------------------|
| Zone                             | Physical/admin assignment  |
| IP address and DHCP reservation  | IT                         |
| Serial number, vendor, model     | SNMP probe                 |
| IPP supported, version, URI      | Get-Printer-Attributes     |
| Duplex, colour, media supported  | IPP or SNMP                |
| SNMP enabled, version, community | Probe                      |
| Page counter present             | Probe                      |
| Vendor print and copy counters   | Vendor documentation, B8.4 |
| Reachable from server directly   | The A6.3 test              |

Devices failing the SNMP row are explicitly marked so the dashboard can show where device-activity tracking is unavailable.

B19.3 Adding a printer

1. Confirm IP and DHCP reservation with IT
2. Create the printer via the admin UI: IP, name, zone
3. Run the capability probe; confirm serial, model and IPP status populate
5. Grant access to the relevant users
6. Print a test page and confirm the job is logged with the correct page count
7. Confirm a walk-up copy at the device produces an entry

B19.4 Backup and restore GAP-07 , blocking

Backed up nightly: a pg\_dump of the database, the uploads directory (subject to retention), and the .env configuration stored separately in the club's secret store.

Restore procedure, tested at go-live and re-tested quarterly:

1. Provision the host, install Node, PostgreSQL, LibreOffice, Ghostscript
2. Restore the dump into an empty database
3. Restore configuration, rotating any secret exposed during the incident

4. Start the application; migrations run automatically and are idempotent
5. Verify /health/ready reports all checks passing
6. Verify job counts against the last known report

RT0 4 hours, RPO 24 hours (NFR-10, NFR-11).

#### B19.5 Rollback

Deployment keeps the previous release directory intact. Rollback is a PM2 switch back plus a service restart. Migrations are forward-only, so a rollback across a migration boundary requires a database restore: any release containing a destructive migration MUST state so in its release note, and there SHOULD NOT be one.

#### B20. Architecture Decision Records

##### ADR-001. IPP as the primary transport, RAW/9100 as fallback

Context. The delivered build sends print jobs over RAW/9100 with hand-built PDL headers. RAW has no feedback channel: a successful send means only that TCP accepted bytes. PDL support varies by manufacturer and cannot be verified per model.

Decision. Implement IPP as the primary transport. Retain RAW/9100 as fallback for devices that fail the IPP probe.

Alternatives rejected. RAW only (status quo): leaves GAP-24 and GAP-25 permanently unclosable, since neither can be answered over RAW. IPP only: excludes older devices for no benefit, since the RAW path already exists and works.

Consequences. Real printer and job state. Verified per-model options. PostScript conversion becomes optional on IPP Everywhere devices. Costs: an IPP client library, a probe and capability cache, and a transport selection path that must distinguish transient failure from unsupported protocol (B6.2). That distinction is the main defect risk.

##### ADR-002. Reject the Windows print spooler and Windows Print Server

Context. The original concept assumed a Windows print stack using the spooler API and WMI. Revisited specifically for the multi-building case.

Decision. Rejected, confirmed.

Reasoning. In favour: real driver-level feature support including finishing and trays, native spooling with retry, and per-job accounting from the PrintService event log, which is better attribution data than a counter delta. Against, and decisive: clients must connect to a print queue, which means driver deployment on every device and directly contradicts the PRD's no-installation requirement; one driver per model on the server; Windows licensing and single-OS lock-in; Point-and-Print is a standing security liability; no visibility into walk-up jobs, which is the hardest requirement; and SMB and RPC across VLANs is considerably more firewall-hostile than a single outbound TCP connection.

Consequences. Two genuine capabilities the spooler would have supplied are built instead: a durable queue with retry (ADR-004) and per-model capability detection (ADR-001). The rejection is only defensible because those are built.

#### ADR-003. Keep Node, Express and pg ; adopt TypeScript incrementally

Context. The department default stack is TypeScript strict, NestJS for structured backends, and Prisma. The delivered system is JavaScript, Express and hand-written parameterised SQL.

Decision. Keep the existing runtime and data-access approach. Adopt TypeScript incrementally with allowJs , typing new modules and converting the pipeline, transport and permission modules first. Do not migrate to NestJS or Prisma.

Reasoning, since this deviates from the default. A rewrite costs months and delivers nothing a user can see, while discarding hand-tuned protocol code that works. Prisma is a poor fit for the two hardest queries in the system: the FOR UPDATE SKIP LOCKED dequeue (B10.4) and the keyset-paginated report queries, both of which would end up as raw SQL anyway. NestJS solves a module organisation problem this codebase does not have; the five-layer structure already provides it.

Consequences. The repository diverges from the department default and this ADR is the stated reason. Type safety arrives gradually rather than at once. The critical modules get typed first, which is where type safety pays.

#### ADR-004. Durable queue in PostgreSQL, not Redis

Context. Print jobs run synchronously inside the HTTP request. Printing volume is low. Conversions can take a minute or more.

Decision. A durable queue in PostgreSQL using FOR UPDATE SKIP LOCKED , with an in-process worker at concurrency 2.

Alternatives rejected. Status quo: a restart loses in-flight jobs, a mid-reboot printer fails permanently, and a long conversion can outlive the browser's patience with no way for the user to learn the outcome. Redis and BullMQ: a second stateful service, a second backup concern and a second failure mode on a single on-premise server, in exchange for throughput this system will never need.

Consequences. Jobs survive restarts and retry automatically. POST /print-file becomes asynchronous ( 202 with a job id), which the frontend must handle by polling job status. That is a real frontend change and it is the cost of this decision.

#### ADR-005. Replace the hand-rolled SNMP client

Context. SNMP is implemented as a hand-written BER/ASN.1 encoder and client, to avoid a dependency.

Decision. Replace with a maintained library. Prefer SNMPv2c, support v3.

Reasoning. The avoided dependency is a binary parser consuming unauthenticated UDP from the network, which is a security surface, and it cannot support SNMPv3. The trade that justified hand-rolling does not survive that observation.

Consequences. One dependency added. Security surface reduced, SNMPv3 becomes possible, and GetBulk reduces poll cost across a large fleet. OID and polling logic port unchanged; only the wire layer is replaced.

#### ADR-006. Serial-number identity

Context. printers.ip\_address is unique and functions as identity. The estate uses DHCP across multiple buildings.

Decision. Identity anchors on serial\_number from SNMP. IP is mutable metadata, unique only among active printers.

Alternatives rejected. Static IPs everywhere: an operational dependency the application cannot enforce, and one wrong reservation reintroduces the bug. MAC address: not readable over IPP, and not reliably readable over SNMP on every model.

Consequences. A changed lease no longer splits the audit trail. A swapped device is detected rather than silently absorbed. Printers that expose no serial fall back to IP identity, and that limitation is recorded per printer rather than assumed away.

#### ADR-007. Site collector deferred from Phase 1

Context. The previous target architecture included a collector to handle genuinely segmented network deployments. The Phase 1 deployment profile does not require that distributed topology.

Decision. Do not deploy or implement the collector in Phase 1. Keep printer communication direct from the core application server. If a future deployment proves that a network boundary requires a collector, introduce it as a separate extension.

Consequences. The Phase 1 system has fewer components, fewer credentials, no offline spool and no collector API. Future extension remains possible because printer transport and monitoring are isolated behind interfaces.

#### ADR-008. Outstanding-impression ledger for walk-up attribution

Context. App-initiated jobs are excluded from walk-up detection by a fixed 45 second window keyed on printer IP.

Decision. Replace the window with a per-printer ledger of outstanding impressions, consumed by counter deltas oldest first, with a generous time cap only to expire failed jobs.

Alternatives rejected. A longer fixed window: widens the interval in which a real walk-up job is wrongly absorbed. A per-printer tuned window: introduces a tuning parameter nobody will maintain.

Consequences. Attribution reconciles against quantity, not elapsed time, so slow jobs are matched correctly and real walk-up jobs during a long print are still caught. Requires an accurate impressions figure per job, which is why jobs.impressions exists (B4.7).

#### ADR-009. Scan-to-folder with a stability gate

Context. Scan detection watches a network share. Files are visible before their write completes.

Decision. Retain scan-to-folder. Add a stability gate: name pattern filter, unchanged size and mtime across two polls, exclusive-open check, then move to processed/ .

Alternatives rejected. Filesystem event notification: unreliable over SMB and does not indicate write completion. Fixed delay before ingestion: guesses at write duration and fails for large scans.

Consequences. No partial ingestion. One extra poll of latency, within NFR-08. Files are moved rather than deleted, because deleting a member's scanned document because it has been counted is not this system's decision.

#### ADR-010. Sandboxed converters

Context. LibreOffice and Ghostscript parse untrusted uploads with the application's privileges.

Decision. Run converters as an unprivileged account with no network access, per-job temporary directories, hard timeouts and resource limits, containerised where the host supports it.

Alternatives rejected. Antivirus alone: does not stop a zero-day against the parser, which is the actual threat. Blocking Office formats: removes a core requirement.

Consequences. A converter compromise is contained rather than a server compromise. Some deployment complexity on a Windows host, where a restricted service account substitutes for a container.

#### ADR-011. Persisted notifications

Context. Notifications are an in-memory FIFO on a single Node process. They are lost on restart and have no per-admin read state.

Decision. Persist to PostgreSQL with a separate per-user read table (B4.8).

Consequences. Notifications survive restarts, each admin has independent read state, and history is queryable. Requires a retention sweep so the table does not grow without bound.

#### ADR-012. Append-only admin audit log

Context. Job activity is logged. Administrative actions, including permission grants, are not.

Decision. Every administrative mutation writes an audit\_log row in the same transaction as the change (INV-07). Append-only, no update or delete path.

Alternatives rejected. Application logs only: not queryable, rotated away, and not tied to the change transactionally. Change-data-capture triggers: opaque to the application and harder to attribute to a human actor.

Consequences. A permission system with a record of permission changes, which is the difference between auditable and merely logged. Small write overhead per mutation. Requires a retention policy (NFR-12).

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